MT Alchemy

NET4207 · Group Project · Flutter + Firebase

COMPLETE PROJECT DOCUMENTATION

MT Alchemy: a human-centred learning and people-development platform

Every page, role, journey and design decision, with the system architecture, data model, security rules, testing and deployment, in one document.

Live site	https://ajaynagarajan2005.github.io/mt-alchemy/
Virtual tour	https://ajaynagarajan2005.github.io/mt-alchemy/tour
Source code	https://github.com/ajaynagarajan2005/mt-alchemy
Backend	Firebase project `mt-alchemy-net4207`: Authentication, Cloud Firestore (asia-southeast1)
Founders	AJAY A.N · MAAHIR FAHEEM · TANUJ SAKTHI KUMAR
Version / date	1.0 · October 2026

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5. **The five user journeys**, step by step
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1. Overview

MT Alchemy is a People Development Consultancy specialising in Emotional Intelligence, Leadership Development, Team Engagement and Workplace Well-being. Its tagline is "*We don't just run activities. We shift mindsets, behaviours, and team culture.*" The platform turns visitors into enquiries, learners and corporate clients.

Four modules in one application

Corporate website

A single landing page whose menu glides to About, Services, Upskilling, Store, Insights and Contact. It adds founders, impact stories, an EQ self-assessment, a "For organisations" enquiry and a cartoon virtual tour reached by QR code.

Learning management system

A course catalogue filtered by each learner's pathway (career stage and interests), purchase and enrolment, a modular player (video, PDF, reading, quiz), progress tracking, and certificates with a public verification link.

E-commerce store

Digital products download after payment; physical products ship with a tracked Processing, Shipped, Delivered timeline. One checkout handles products, courses and employer seats. Payments run in test mode.

Role-based portals

Five roles (Guest, Individual, Employer, Trainer, Super Admin), each with its own navigation, notifications and permissions enforced by database security rules.

The project in numbers

5 user roles	5 user journeys	12 courses	6 products	22 Firestore collections	47 routes
125 Dart files	24k lines of Dart	56 automated tests	39 live security checks	30 photographs	6 videos

How the platform meets the Employer Project brief

Brief requirement	Where it is delivered
Five core journeys (A to E)	Journey A learning to certificate; B digital product to download; C physical product to delivery; D employer seats, assignment and progress; E corporate enquiry to customised programme. See chapter 5.
Five roles with authentication	Guest, Individual, Employer, Trainer, Super Admin. Firebase Authentication plus role-based redirects in the app and security rules in the database.

CRUD for courses, products and users	Courses (trainer and admin, with lessons and quizzes), products (admin), users (admin: create, change role, disable, delete), plus articles, impact stories, founders, employees and assignments.
Course player: video, PDF, quiz, progress, Previous / Mark complete / Next	Modular player with a progress percentage, per-lesson views, quiz scoring (70% pass mark) and persisted progress.
Responsive on web, tablet and mobile	Three breakpoints (600 px and 1024 px), fluid gutters, drawers below 1024 px. Verified at 1440 px and 375 px.
Backend: Firebase (Auth, Firestore, Storage) or Supabase	Authentication and Firestore are used. Storage is not used (see chapter 11); files are served from the site itself.
Provider / Riverpod / Bloc	Provider with one `ChangeNotifier` store per domain.
README, ER diagram, maintenance guide	`README.md`, `docs/ER_DIAGRAM.md`, `docs/MAINTENANCE.md`, and this document.
Innovation	EQ self-assessment with course recommendation, employer analytics charts, Impact Stories carousel, learning pathways, admin free-access tools, public certificate verification, dark mode, cartoon virtual tour with QR code.
Scaffold, Container, Form (prototype brief)	Two Scaffolds (public `AppShell`, portal `PortalShell`) with AppBars; Containers for cards, padding, decoration and grouping; ten Forms with required-field validation. Code is annotated with `SCAFFOLD:`, `CONTAINER (` and `FORM:` comments.

2. Information architecture

The site has one public area and four signed-in portals. The menu (About, Services, Upskilling, Store, Insights, Contact) is available everywhere; on the home page, each item glides to its section of the landing page, and from any other page it opens the full page.

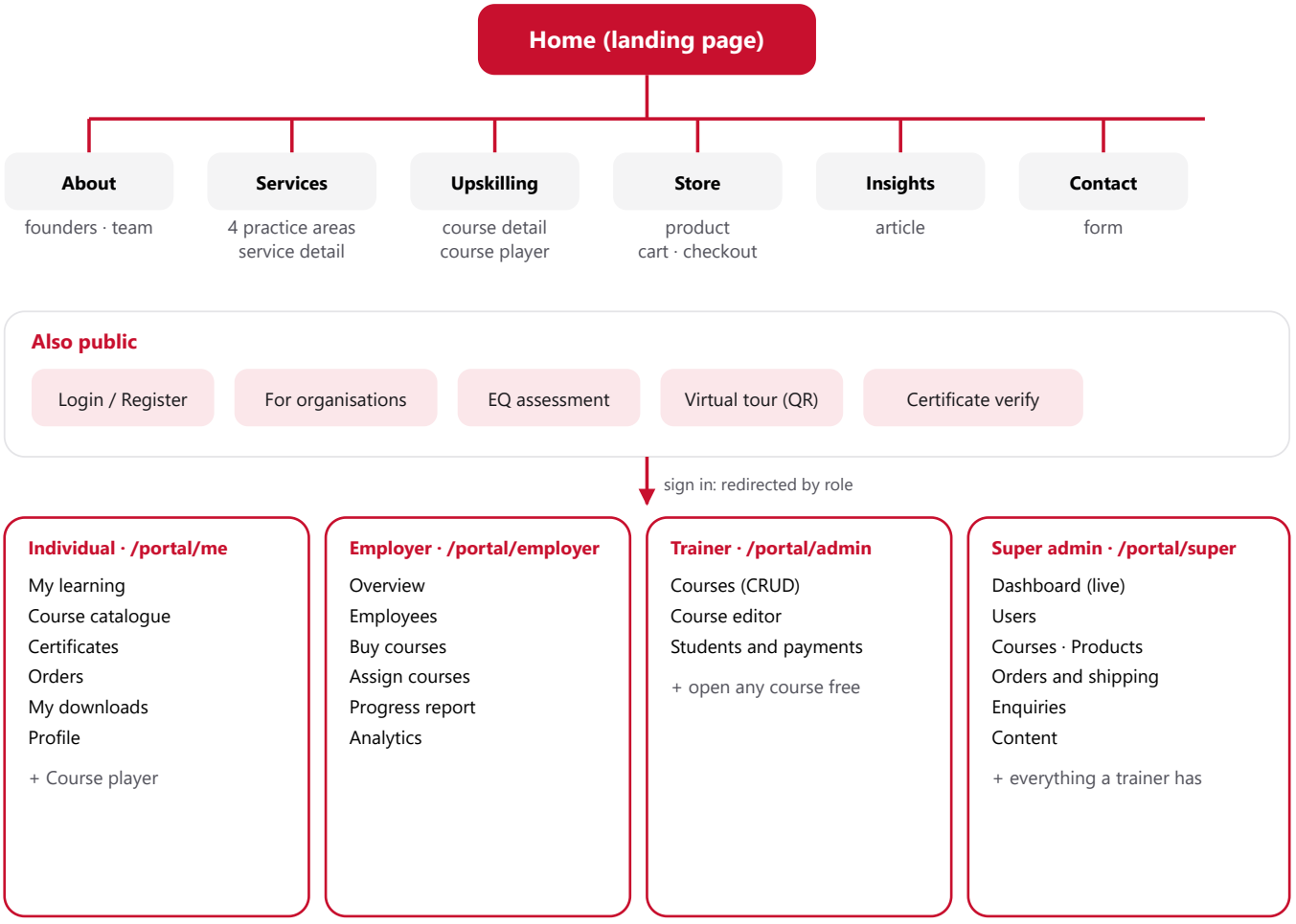


Figure 2.1. Sitemap. Portals are reachable only by the matching role; any other role is redirected to its own portal home.

3. System architecture

The application is a single Flutter codebase that talks directly to Firebase. There is no custom server: business rules run in the client stores, and Firestore security rules decide who may read or write each document. Everything is delivered as a static web app (GitHub Pages) that connects to Firebase over HTTPS.

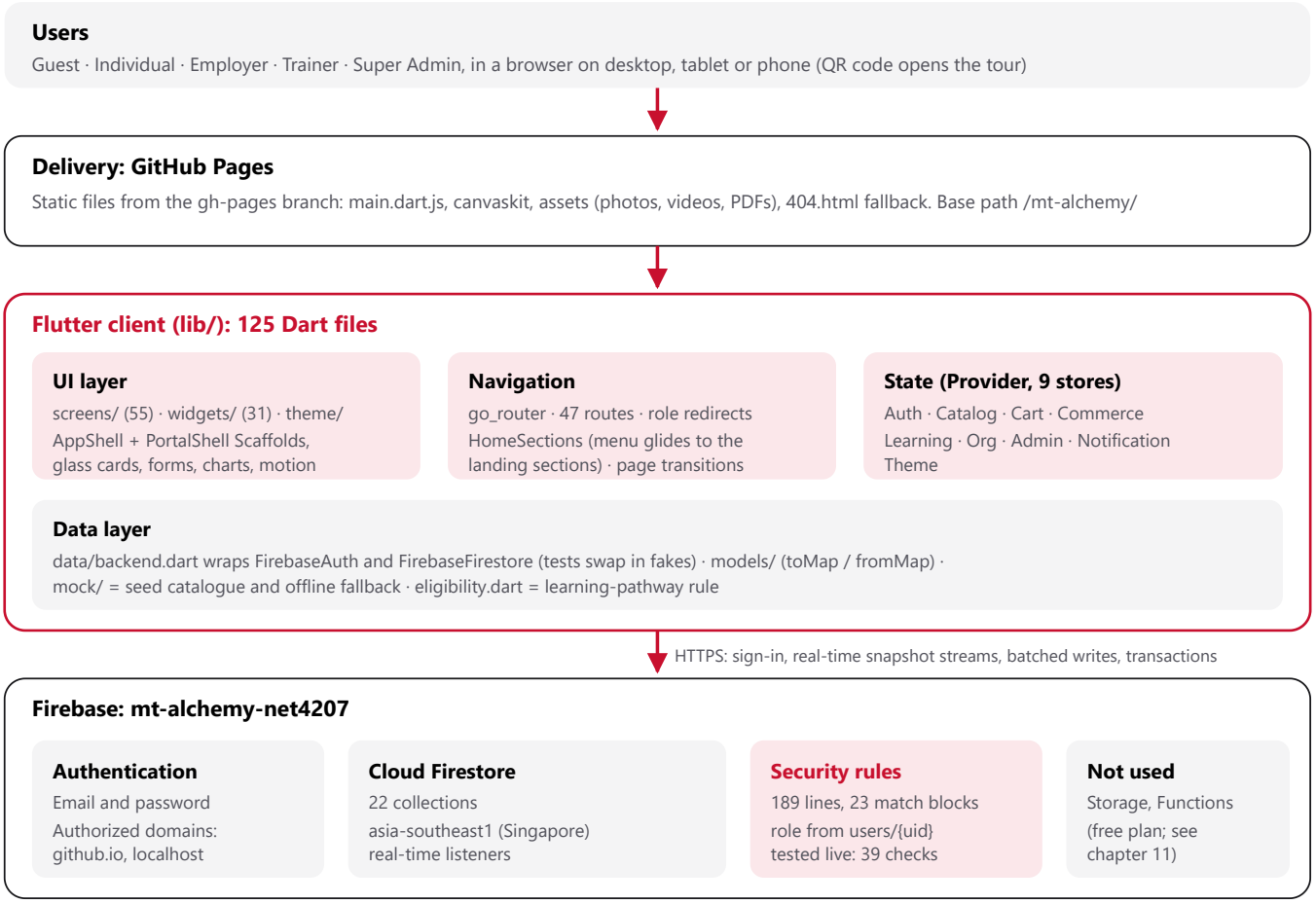


Figure 3.1. Layers of the platform. Firestore pushes changes back over live listeners, so every screen updates in real time.

Technology

Area	Choice	Why
UI framework	Flutter 3.47 (Dart 3.13), Material 3 with a custom theme	One codebase for web, Android and iOS; rich animation.
Navigation	`go_router` 18	Declarative routes, clean URLs, redirect guards per role.
State	`provider` with `ChangeNotifier` stores	Brief-approved; simple and testable.
Auth and data	`firebase_auth`, `cloud_firestore`	Managed sign-in and a real-time database with declarative security rules.
Charts	`fl_chart`	Employer analytics (donut, bars, line).
Media	`video_player`, `flutter_3d_controller`, `flutter_animate`	Looping films, a 3D crystal, staggered reveals.

Utilities	`qr_flutter`, `file_picker`, `image`, `url_launcher`, `intl`, `google_fonts`, `hugeicons`	QR code, founder photo upload and resize, downloads, formatting, typography, icons.
Video production	Remotion (React) renders to H.264	Programmatic, repeatable films: landing loop, brand film, cartoon tour.
Hosting	GitHub Pages (static) + Firebase (backend)	Free tier; the web app needs no server.

Code structure

Folder	What lives there
`lib/data/`	`backend.dart` (Firebase access, friendly error messages), `models/` (Course with lessons and quiz questions, Product, AppUser, Organisation, Employee, Article, Review, Founder, CartItem), `mock/` (seed catalogue), `eligibility.dart`
`lib/state/`	Nine `ChangeNotifier` stores. Each subscribes to the signed-in user's Firestore queries and exposes getters plus async actions that return an error message or null.
`lib/router/`	`routes.dart` (every path), `app_router.dart` (shell routes, role guards, fade transitions), `home_sections.dart` (landing anchors)
`lib/screens/`	55 files: public pages, `learn/` (player and lessons), `portal/` (learner, employer, trainer, super admin), `tour/`, `certificate/`, `assessment/`
`lib/widgets/`	31 reusable widgets: GlassPanel, AlchemyButton, AnimatedCard, CardImage, CourseThumb, Reveal, TiltCard, ReviewsSection, PathwayFields, the two layout shells, the notification bell
`lib/theme/`	Colour, typography, spacing, radius and motion tokens for light and dark
`test/`	13 test files on a fake Firestore and mock Auth (`helpers/fake_backend.dart`)
`tools/`	`seed/` (seed.mjs, verify.mjs), `export_seed.dart`, `make_pdfs.py`, `serve_web.mjs`, `deploy_pages.sh`
`video/`	Remotion project: HeroLoop, ServicesLoop, BrandIntro, SiteTour
`web/`, `assets/`	HTML shell and downloads (`web/downloads/*.pdf`); 30 photographs, 6 videos, a 3D model, textures

How data flows

1. **Sign-in.** `AuthStore` listens to Firebase Auth. When a user signs in it streams `users/{uid}` (role, organisation, pathway) and their organisation document.
2. **Fan-out.** The other stores listen to `AuthStore`. When the uid changes they cancel old listeners and subscribe to that user's data, for example `enrolments where userId == uid`.
3. **Rendering.** Screens `watch` stores and rebuild automatically. `CatalogStore` starts with the bundled catalogue so pages paint instantly and work offline; Firestore then replaces it live.
4. **Writes.** Checkout, assignment and enrolment use batched writes or transactions (stock decrement). Lesson progress updates optimistically, so the interface responds before the server confirms.
5. **Errors.** Every action returns a friendly message (for example "That email and password do not match an account.") instead of throwing.

Routing and role guards

Public pages share `AppShell`: one Scaffold with a glass AppBar and an end drawer below 1024 px. Signed-in pages share `PortalShell`: a second Scaffold with its own AppBar, a side rail (drawer on small screens), the notification bell, a theme toggle and one scroll area with a visible scrollbar. All portal pages are flat sibling routes, because nested child routes would stack a page underneath that Flutter never lays out.

Role	Home after sign-in	Guard
Guest	public site	Portal paths and `/learn` redirect to `/login?from=…`, and return afterwards.
Individual	`/portal/me`	May open only `/portal/me/*`.
Employer	`/portal/employer`	May open only `/portal/employer/*`.
Trainer	`/portal/admin`	May open only `/portal/admin/*`.
Super admin	`/portal/super`	May open `/portal/super/*` and the trainer's course editor and students pages.

The router waits for Firebase to restore the session before deciding, so a signed-in user is never bounced to the login screen on refresh.

Theme switching

Colour tokens are static getters on the active palette. When the brightness changes, every element is marked for rebuild so each widget re-reads the tokens, without recreating the tree; the current page, scroll position and widget state survive. The toggle reads the live brightness at press time, so it always flips.

Build and delivery pipeline

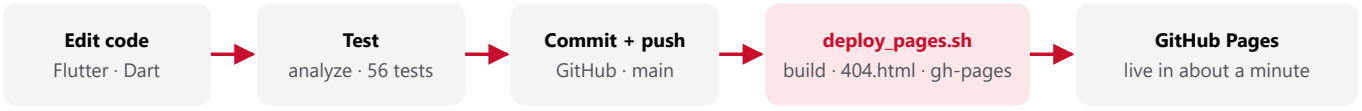


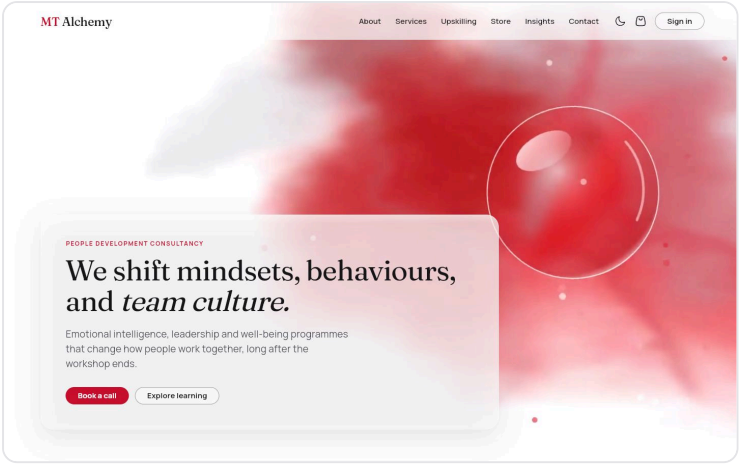
Figure 3.2. From a code change to the live site. Rules and data changes go to Firebase separately (chapter 10).

4. Page-by-page guide

Each entry shows the page as it appears on the live site, its route, who can open it, what it is for, the Firestore data it uses, and what a user can do. Screenshots were captured from the published site at 1440 px.

4.1 Public website

Open to everyone. The home page is one long landing page; its menu items glide to the matching section.



1. Home: landing hero

`/`

Everyone

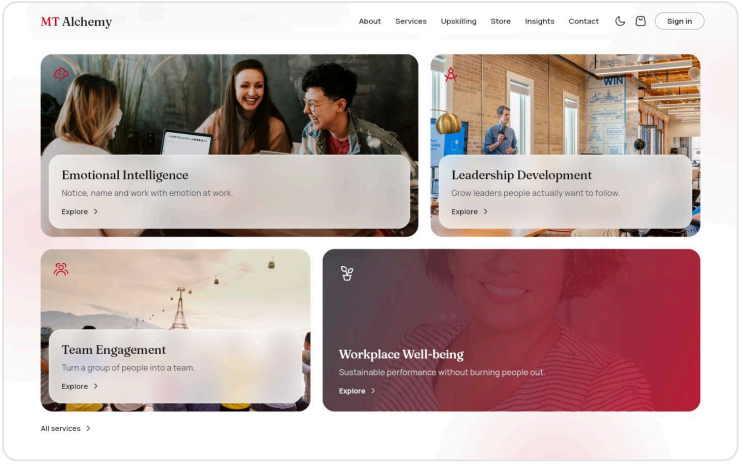
The first impression: a looping ink-in-water film with a floating glass crystal behind glass copy, "We shift mindsets, behaviours, and team culture", and two calls to action.

DATA

None (static hero).

WHAT YOU CAN DO

- Book a call, Explore learning
- Menu and theme toggle
- Cart badge shows the item count



2. Home: services

`/ (section)`

Everyone

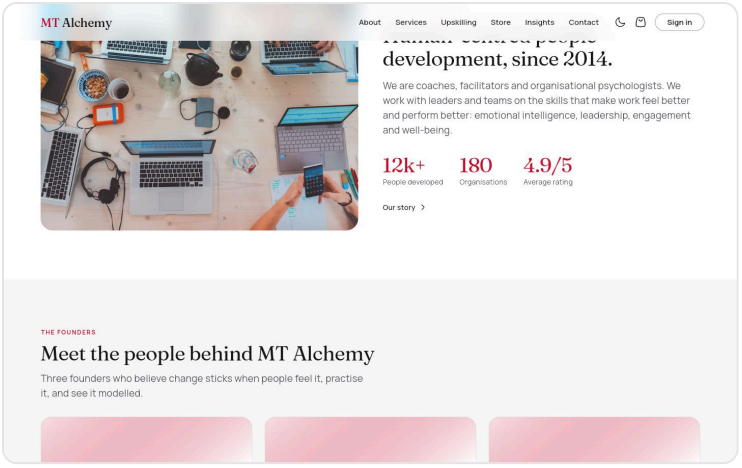
The hero zooms into four photo cards, one per practice area, with a 3D tilt on hover.

DATA

Services (bundled).

WHAT YOU CAN DO

- Open a service
- All services link
- Menu "Services" glides here



3. Home: about

`/ (section)`

Everyone

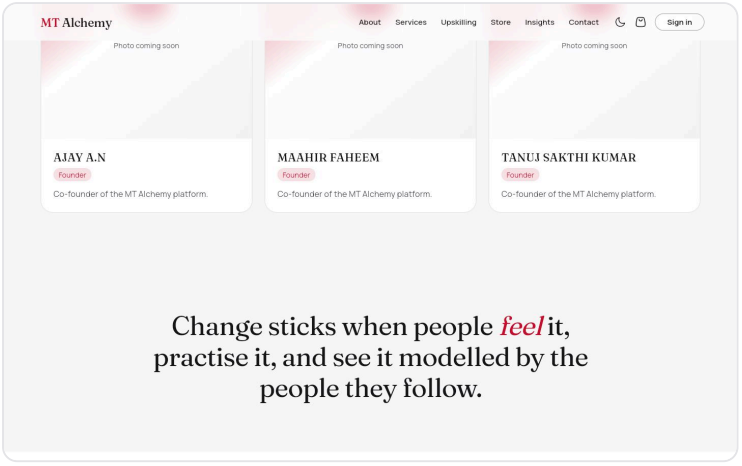
Who we are: a team photograph, a short introduction and three key numbers. The founders section begins underneath.

DATA

None (static content).

WHAT YOU CAN DO

- Our story link
- Menu "About" glides here



4. Home: founders and manifesto

`/ (section)`

Everyone

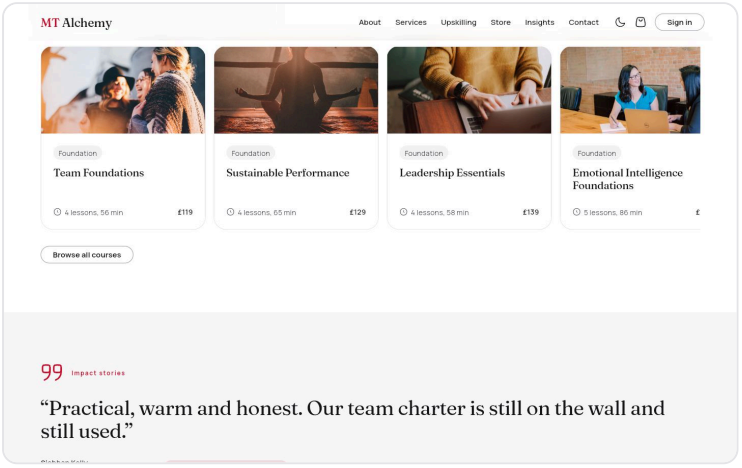
Three animated founder cards (AJAY A.N, MAAHIR FAHEEM, TANUJ SAKTHI KUMAR) with a photo slot each, followed by the statement of belief. A placeholder with initials shows until a photo is added.

DATA

founders (live).

WHAT YOU CAN DO

- Hover: the card lifts
- Super admin: Upload or Change photo, Edit details



5. Home: courses and impact stories

`/ (section)`

Everyone

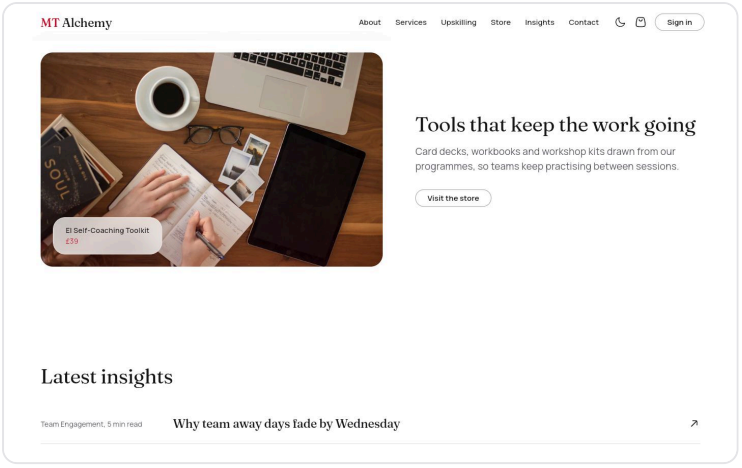
A snapping carousel of featured courses with photos, then client quotes with a headline result that rotate every seven seconds, with dots to jump between stories.

DATA

courses (published only), impactStories (live).

WHAT YOU CAN DO

- Open a course
- Browse all courses
- Jump to a story
- No auto-rotation under reduced motion



6. Home: store and insights

`/ (section)`

Everyone

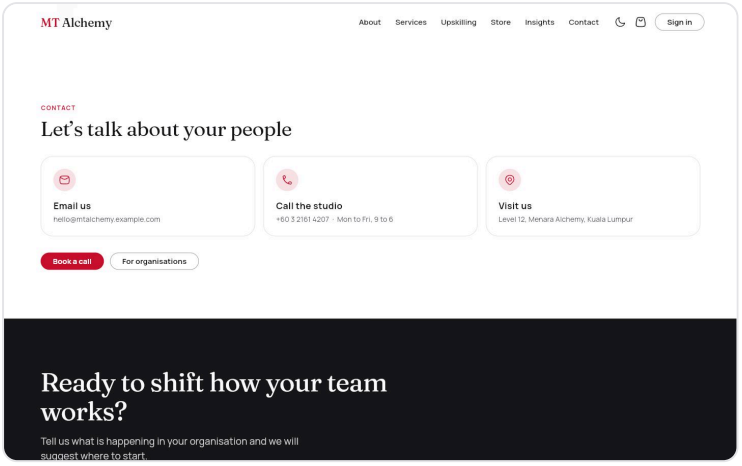
A featured product from the live catalogue and the three latest articles as editorial rows.

DATA

products, articles (live).

WHAT YOU CAN DO

- Visit the store
- Open an article



7. Home: contact and call to action

`/ (section)`

Everyone

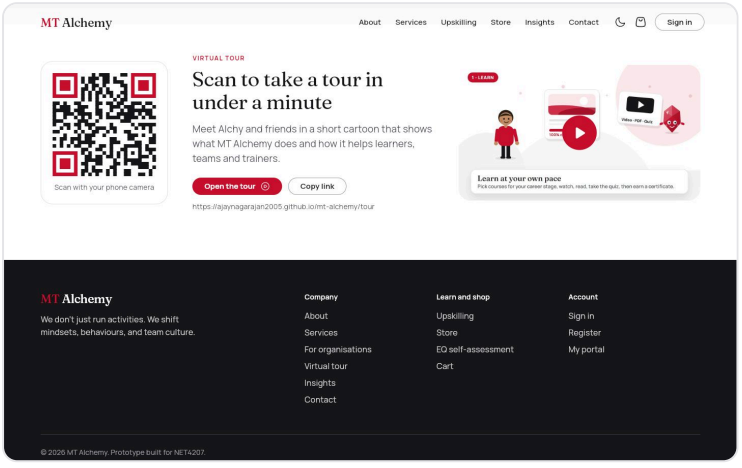
Three animated contact cards and a closing band inviting organisations to design a programme.

DATA

None.

WHAT YOU CAN DO

- Book a call
- For organisations
- Register your organisation



8. Home: virtual tour QR

`/ (final section)`

Everyone

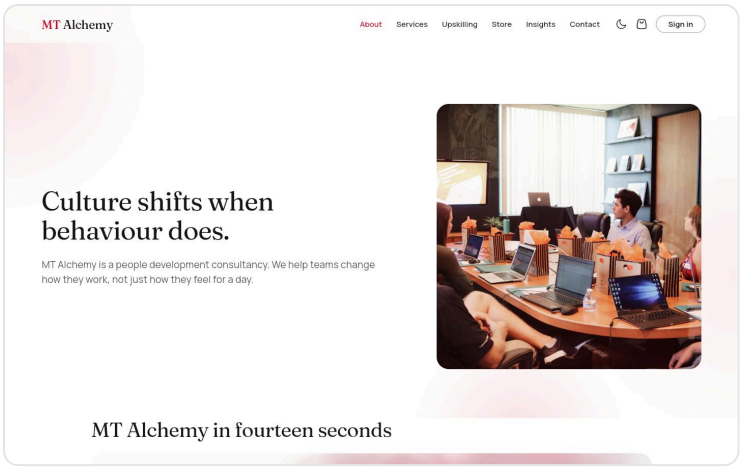
The last section: a QR code that opens the cartoon tour, with the address, copy and open buttons, and a clickable preview of the film.

DATA

None. The address is computed from the host.

WHAT YOU CAN DO

- Scan or open the tour
- Copy link



9. About

`/about`

Everyone

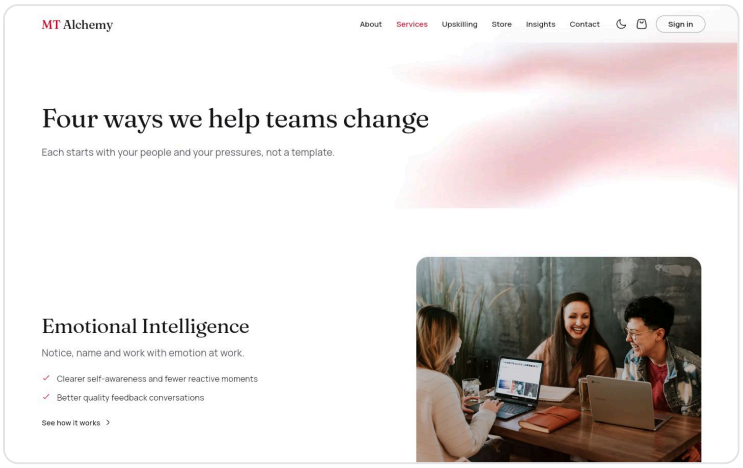
The story of MT Alchemy: hero, the brand film, guiding principles, the founders and the team.

DATA

founders (live).

WHAT YOU CAN DO

- Play the brand film
- Read the principles
- Book a call



10. Services

`/services`

Everyone

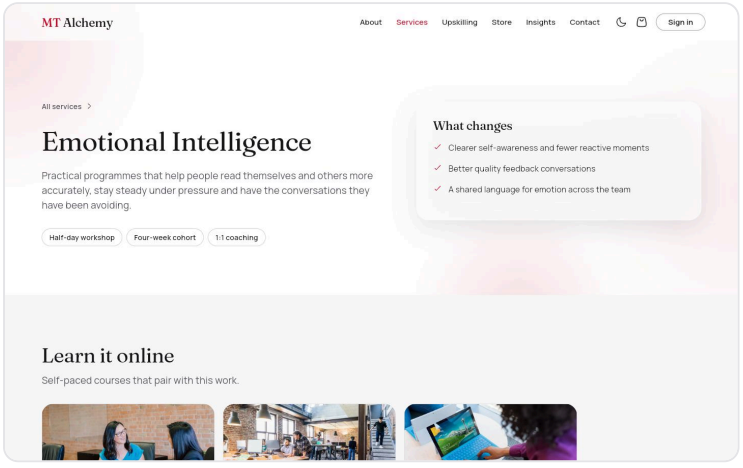
The four practice areas: Emotional Intelligence, Leadership Development, Team Engagement and Workplace Well-being, each with a photograph and a one-line promise.

DATA

Services (bundled).

WHAT YOU CAN DO

- Open a service



11. Service detail

`/services/:slug`

Everyone

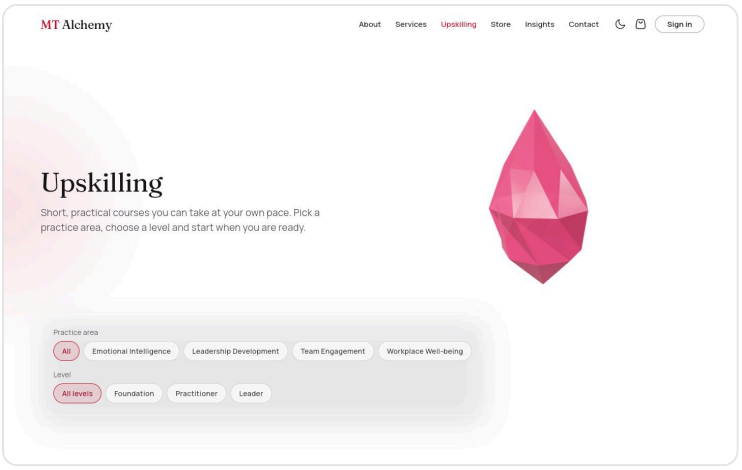
One practice area in depth: summary, what changes, formats, related self-paced courses and a client quote.

DATA

courses, impactStories (live).

WHAT YOU CAN DO

- Open a related course
- Book a call



12. Upskilling (course library)

`/upskilling`

Everyone

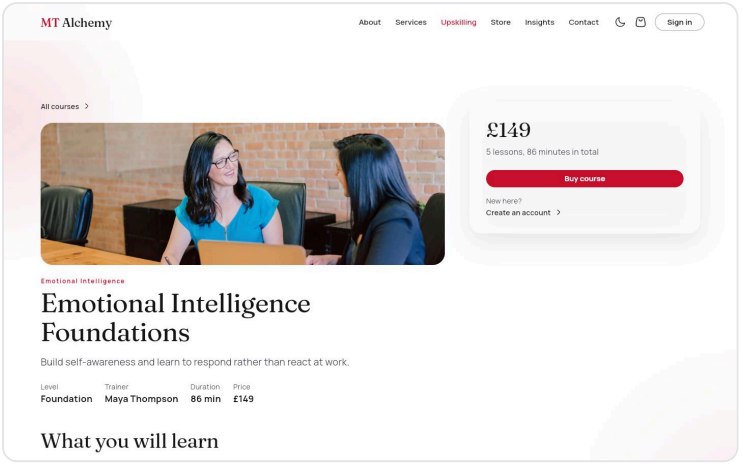
The public course library with level and practice-area filters and photo cards that lift and zoom on hover.

DATA

courses (live).

WHAT YOU CAN DO

- Filter by level or area
- Open a course



13. Course detail

`/upskilling/:id`

Everyone (actions depend on role)

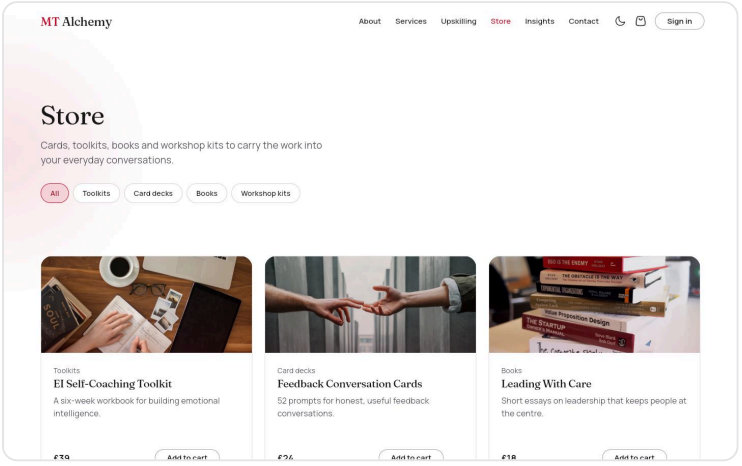
Photo header, what you will learn, lessons (type and duration), reviews and a summary panel whose button depends on who is looking.

DATA

courses, reviews, enrolments, assignments.

WHAT YOU CAN DO

- Guest: sign in or create account
- Learner: Buy course (if in pathway) or See courses in your pathway
- Employee: Start assigned course
- Employer: buy seats
- Staff: Open free, Students, Edit
- Learners post reviews



14. Store

`/store`

Everyone

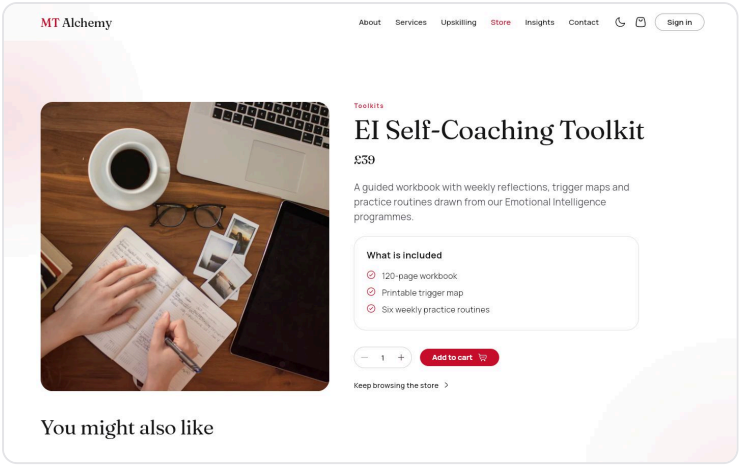
Toolkits, card decks, books and workshop kits with a category filter and photo cards.

DATA

products (live).

WHAT YOU CAN DO

- Filter by category
- Open a product
- Add to cart



15. Product

`/store/:id`

Everyone

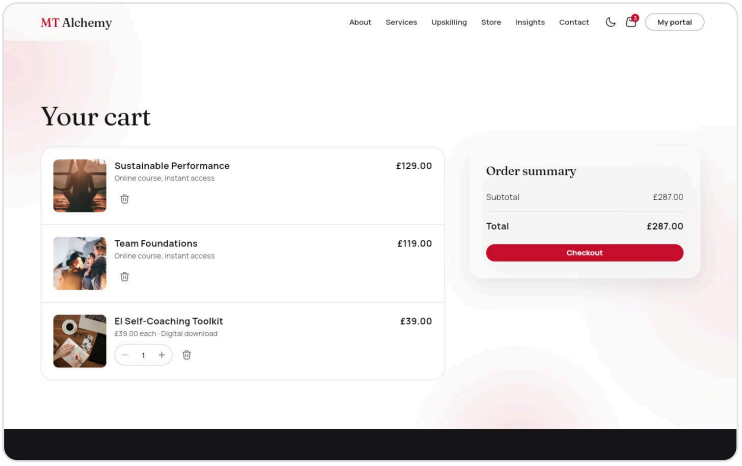
Product photo, description, what is included, price, digital or shipped, reviews and related products.

DATA

products, reviews.

WHAT YOU CAN DO

- Add to cart
- Quantity
- Read reviews



16. Cart

`/cart`

Everyone

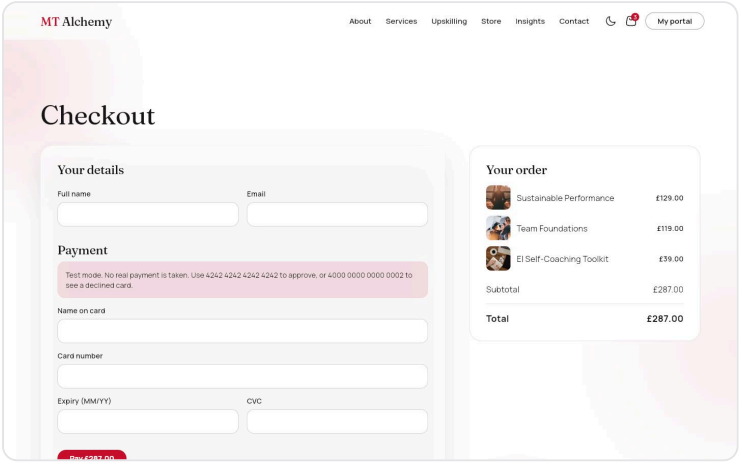
Mixed lines (products, a course, employer seats) with quantity steppers, remove, totals and a free-delivery nudge. A guest cart is merged into the saved cart on sign-in.

DATA

carts/{uid} when signed in.

WHAT YOU CAN DO

- Change quantity
- Remove
- Go to checkout



17. Checkout

`/checkout`

Signed in

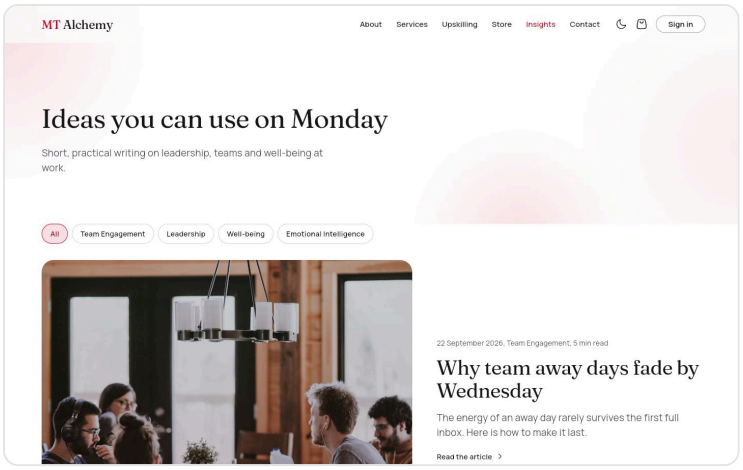
Details, a delivery address only when something ships, and a test-mode card form. The confirmation page (`/checkout/success`) offers the next step: Start learning, Assign your seats, My downloads or Track my order.

DATA

orders, payments, shipping, downloads, enrolments, organizations (seats), products (stock).

WHAT YOU CAN DO

- Pay (validated card number, expiry, CVC)
- Declined test card shows an error



18. Insights

`/insights`

Everyone

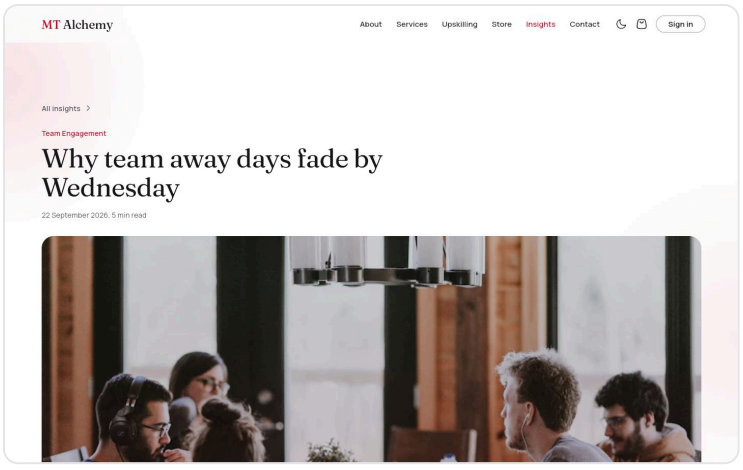
Short practical articles with a category filter and a featured story.

DATA

articles (live).

WHAT YOU CAN DO

- Filter by category
- Open an article



19. Article

`/insights/:slug`

Everyone

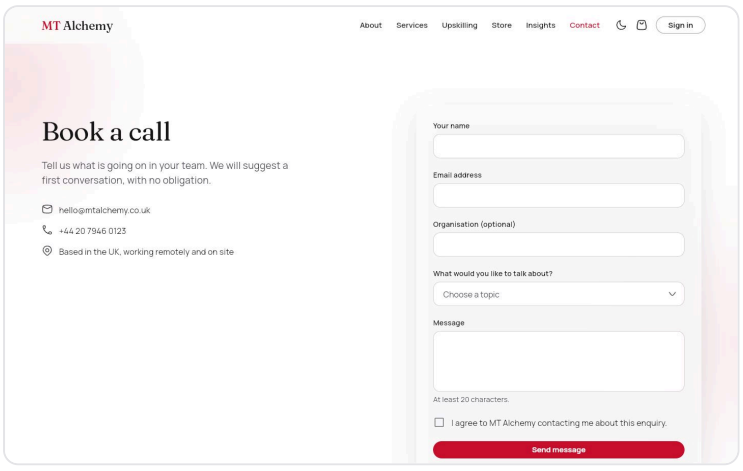
A full article with a photo header, reading time and related articles.

DATA

articles.

WHAT YOU CAN DO

- Read
- Open a related article



20. Contact

`/contact`

Everyone

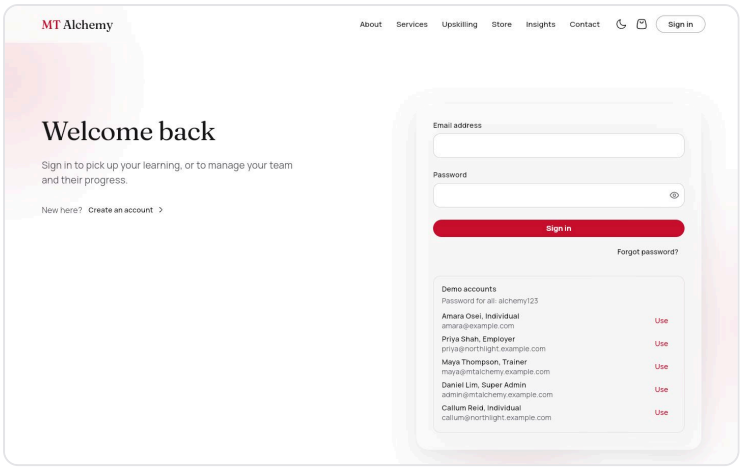
A form (name, email, topic, message, consent) with validation that saves to Firestore and shows a success state.

DATA

contactMessages (write).

WHAT YOU CAN DO

- Send a message
- Send another



21. Sign in

`/login`

Guests

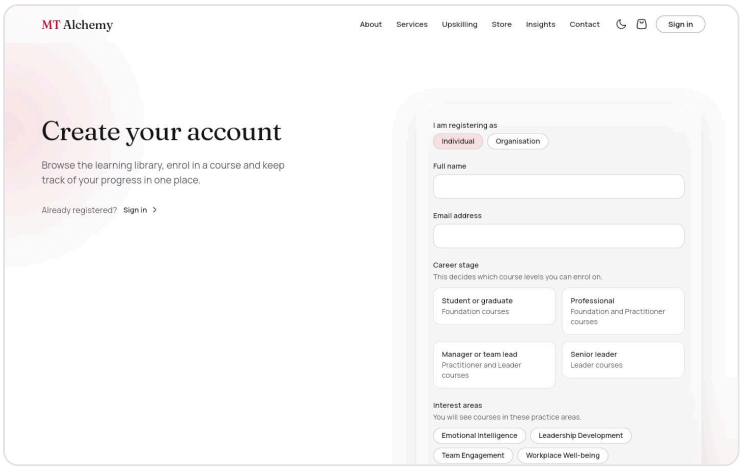
Email and password with validation, "Forgot password?" (sends a Firebase reset email) and one-tap demo accounts.

DATA

Firebase Auth; users.

WHAT YOU CAN DO

- Sign in and land on your portal
- Reset password
- Use a demo account



22. Register: individual

`/register`

Guests

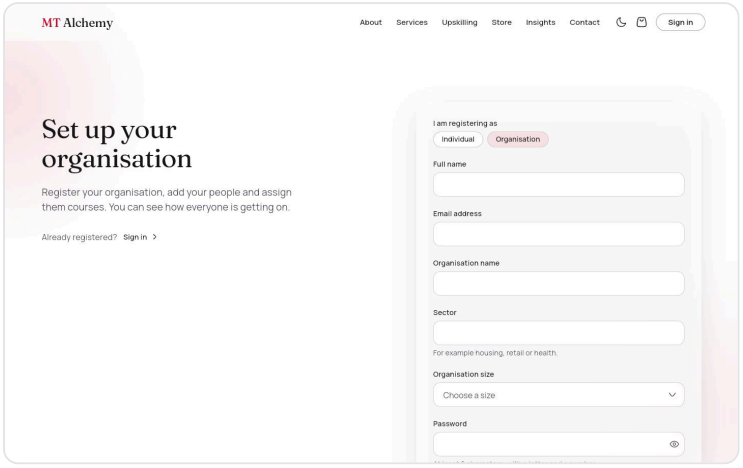
Name, email, password, gender (radio), country (dropdown), hobbies (checkboxes) and the **learning pathway**: career stage plus interest areas, all validated.

DATA

users (create).

WHAT YOU CAN DO

- Create account, then go to the course catalogue



23. Register: organisation

`/register?type=organisation`

Guests

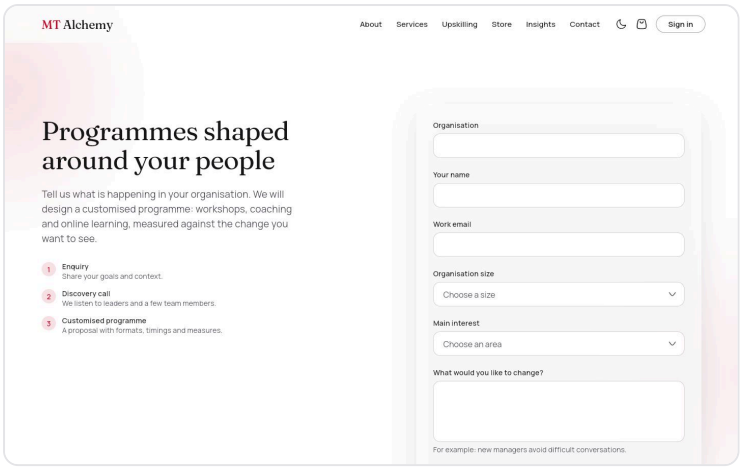
Journey D starts here: organisation name, sector and size, plus the employer's own login.

DATA

users, organizations (create).

WHAT YOU CAN DO

- Register, then go to the employer portal



24. For organisations (Journey E)

`/organisations`

Everyone

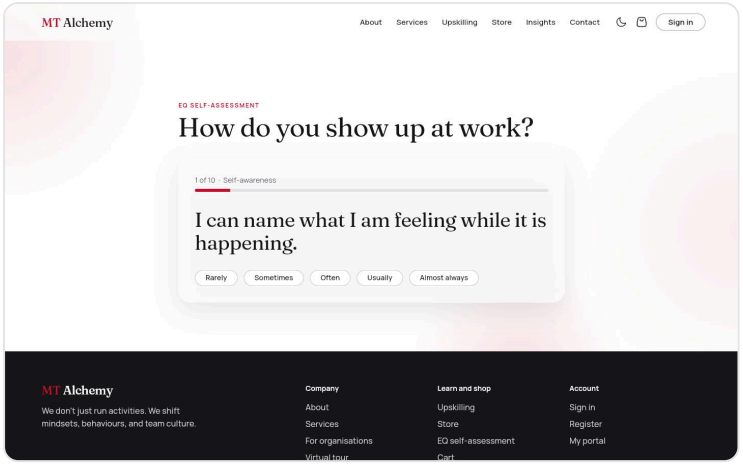
A corporate enquiry form with a three-step explanation: enquiry, discovery call, customised programme.

DATA

enquiries (create).

WHAT YOU CAN DO

- Send enquiry
- Then register the organisation



25. EQ self-assessment

`/eq-assessment`

Everyone (results saved if signed in)

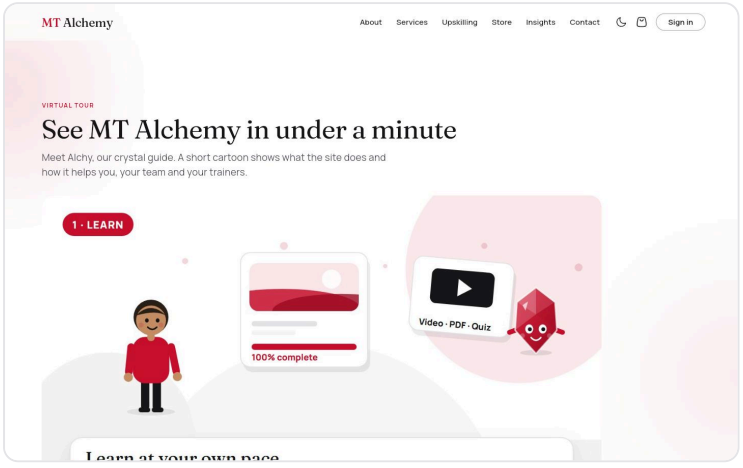
Ten statements across five emotional-intelligence areas, one at a time. The result shows animated bars, a strength, a growth edge and a recommended course.

DATA

assessments (write when signed in).

WHAT YOU CAN DO

- Answer, go back, retake
- View the recommended course



26. Virtual tour

`/tour`

Everyone

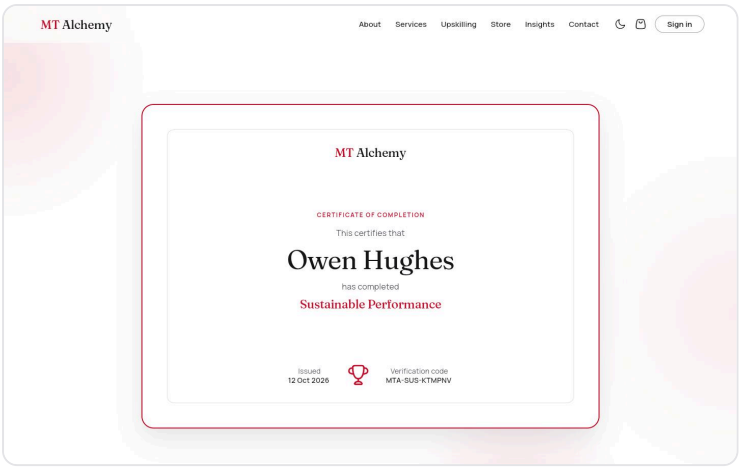
The page the QR code opens: the cartoon explainer film, then six "Try it yourself" cards linking to learning, shop, teams, trainers, the EQ check and insights.

DATA

None.

WHAT YOU CAN DO

- Play the film
- Open any part of the site
- Create a free account



27. Certificate

`/certificates/:code`

Everyone (public by design)

A printable certificate with the learner's name, course, issue date and verification code. Anyone with the link can verify it, and an unknown code shows "Certificate not found".

DATA

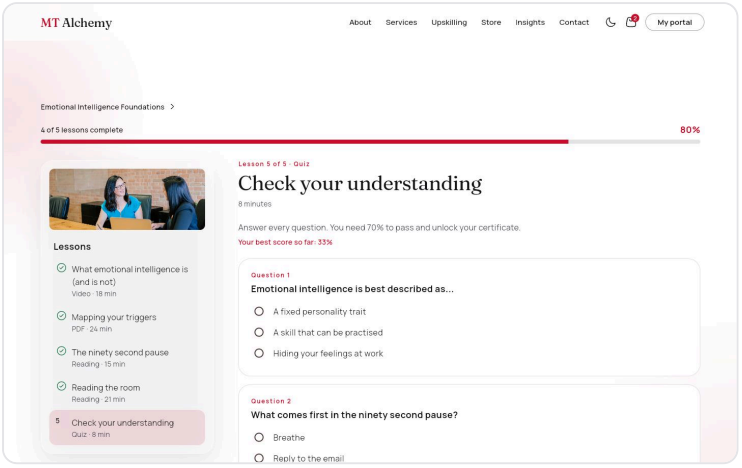
certificates (public read).

WHAT YOU CAN DO

- Share the link
- Verify a code

4.2 Course player

Where learning happens. Signed-in learners open the courses they own; staff can open any course free.



28. Course player: video lesson

`/learn/:id`

Enrolled learners; staff (free preview)

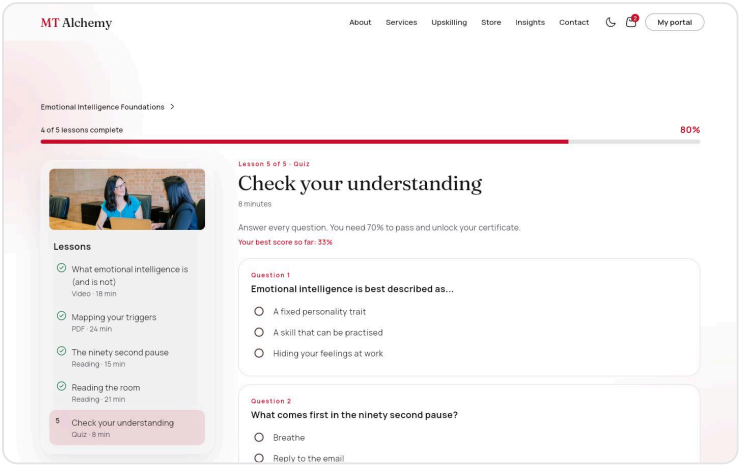
A progress bar with a percentage, a lesson list with a photo banner and a type and duration for each lesson, and the lesson view. Video lessons use the film player; PDF lessons open the workbook; readings show the text.

DATA

courses, enrolments, certificates.

WHAT YOU CAN DO

- Previous, Mark as complete, Next
- Progress is saved to Firestore
- Finishing creates the certificate



29. Course player: quiz lesson

`/learn/:id`

Enrolled learners; staff (free preview)

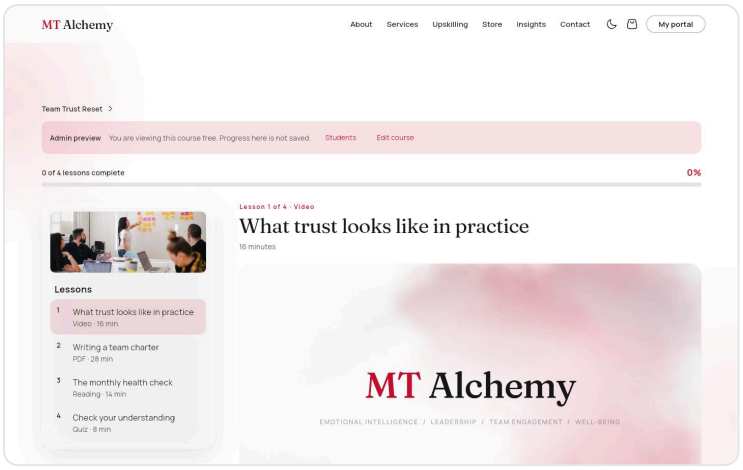
One choice per question. Submitting shows which answers were right; 70% passes and completes the lesson, otherwise "Try again".

DATA

enrolments.quizScores.

WHAT YOU CAN DO

- Answer all questions
- Submit
- Retry until passed



30. Admin preview

`/learn/:id`

Trainer, super admin

Staff open any course free. A banner says "You are viewing this course free. Progress here is not saved" and offers Students and Edit course.

DATA

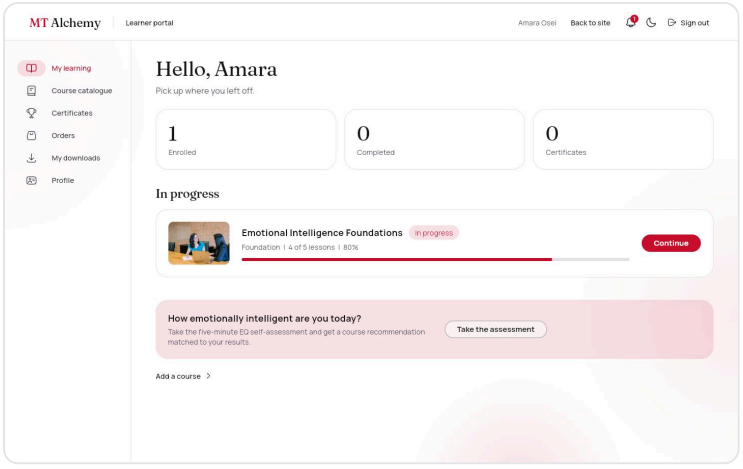
courses.

WHAT YOU CAN DO

- Try every lesson
- Jump to Students or the editor

4.3 Learner portal

For individuals. Side navigation: My learning, Course catalogue, Certificates, Orders, My downloads, Profile. The top bar has a notification bell, a theme toggle and sign out.



31. My learning

`/portal/me`

Individual

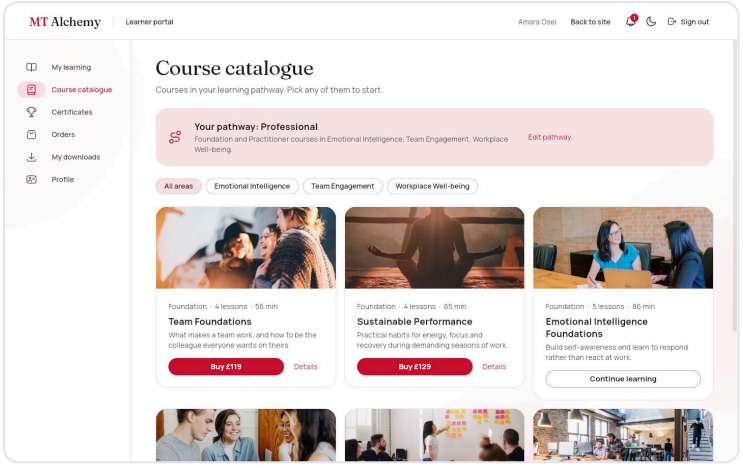
Totals for enrolled, completed and certificates; courses assigned by an employer; courses in progress with a photo and progress bar; completed courses; and a prompt to take the EQ check.

DATA

enrolments, assignments, certificates, courses.

WHAT YOU CAN DO

- Continue or Review a course
- Start an assigned course
- Add a course



32. Course catalogue ("Add a course")

`/portal/me/courses`

Individual

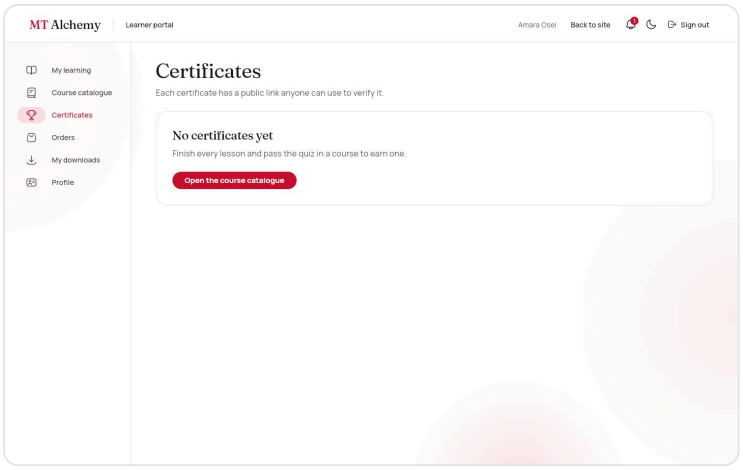
Only courses in the learner's pathway, with a banner showing the pathway and area chips. Courses outside it are listed as locked with the reason.

DATA

courses, users.pathway.

WHAT YOU CAN DO

- Buy a course
- Details
- Edit pathway
- Show locked courses



33. Certificates

`/portal/me/certificates`

Individual

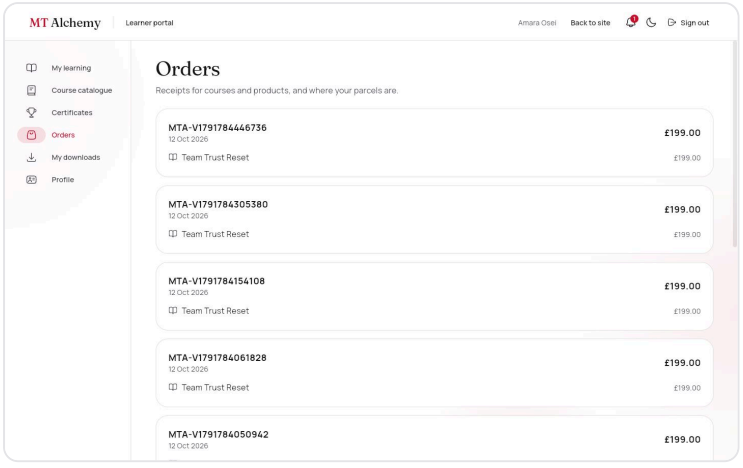
Every earned certificate with its course photo and code.

DATA

certificates.

WHAT YOU CAN DO

- View a certificate



34. Orders

`/portal/me/orders`

Individual

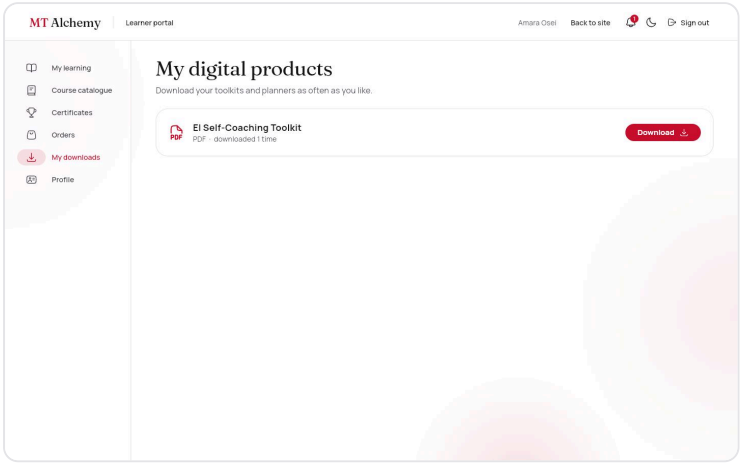
Receipts for courses and products. Physical orders show a Processing, Shipped, Delivered timeline with the tracking number.

DATA

orders, shipping.

WHAT YOU CAN DO

- See delivery progress



35. My downloads

`/portal/me/downloads`

Individual

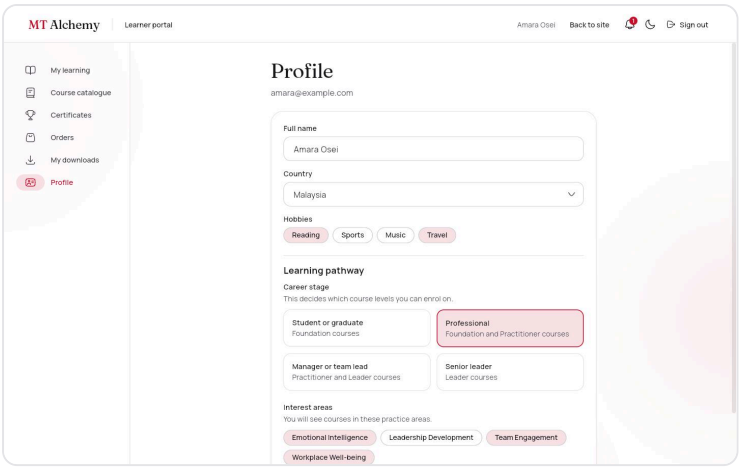
Digital products the learner owns, with a download count.

DATA

downloads.

WHAT YOU CAN DO

- Download (the count goes up)



36. Profile

`/portal/me/profile`

Individual

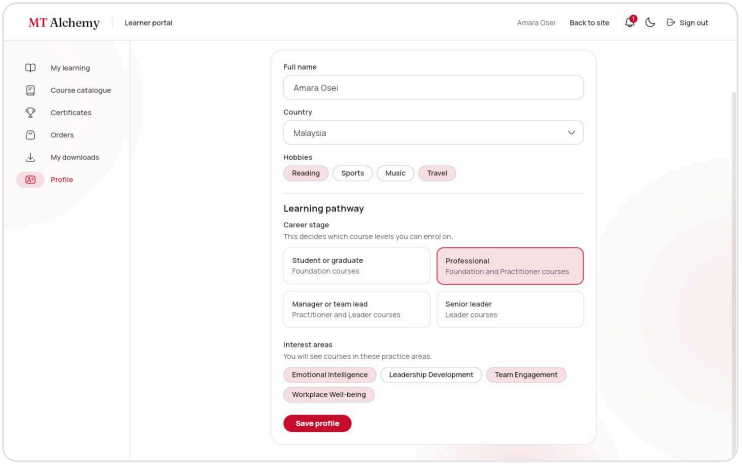
Edit name, country and hobbies.

DATA

users.

WHAT YOU CAN DO

- Save profile



37. Profile: learning pathway

`/portal/me/profile`

Individual

The same pathway control used at registration; changing it changes which courses are open.

DATA

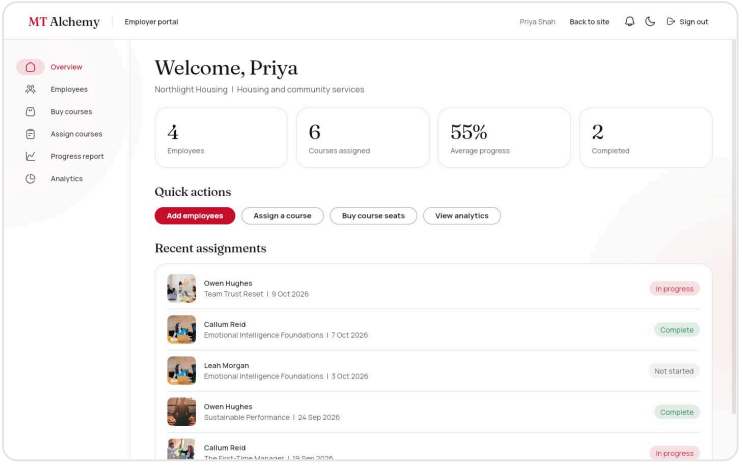
users.careerStage, interests.

WHAT YOU CAN DO

- Choose a career stage
- Choose interest areas
- Save

4.4 Employer portal

For organisations (Journey D). Side navigation: Overview, Employees, Buy courses, Assign courses, Progress report, Analytics.



38. Overview

`/portal/employer`

Employer

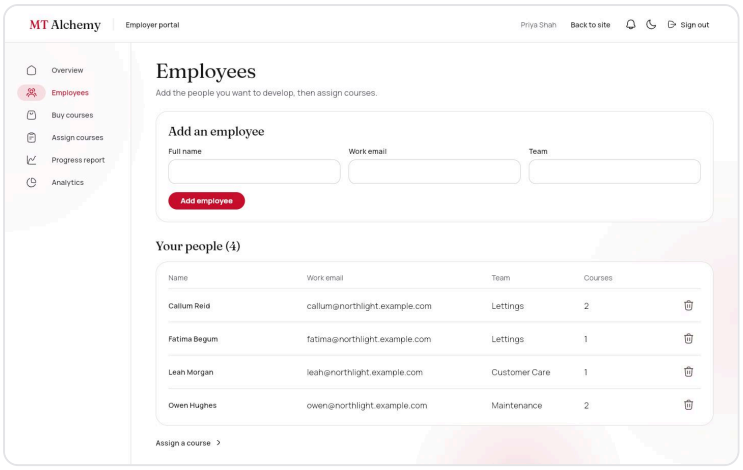
Employees, courses assigned, average progress and completions, quick actions, and the five most recent assignments with course photos.

DATA

employees, assignments, enrolments.

WHAT YOU CAN DO

- Add employees
- Assign a course
- Buy course seats
- View analytics



39. Employees

`/portal/employer/employees`

Employer

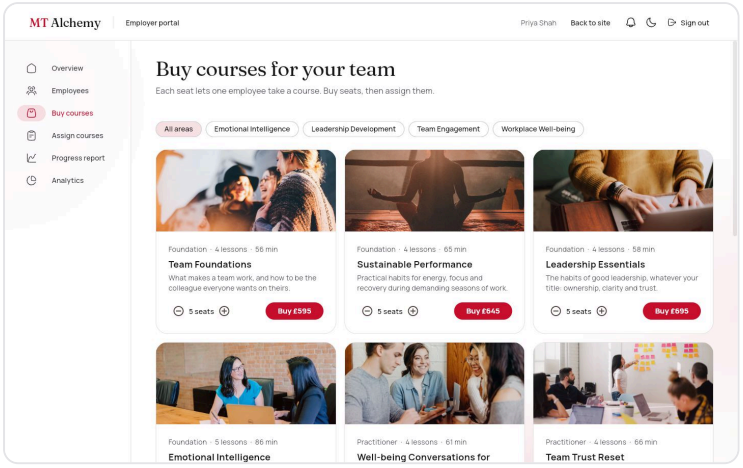
A form to add people (name, email, team) and the list with their number of courses. People join the organisation automatically when they register with that email.

DATA

employees.

WHAT YOU CAN DO

- Add
- Remove (with confirmation)



40. Buy courses

`/portal/employer/courses`

Employer

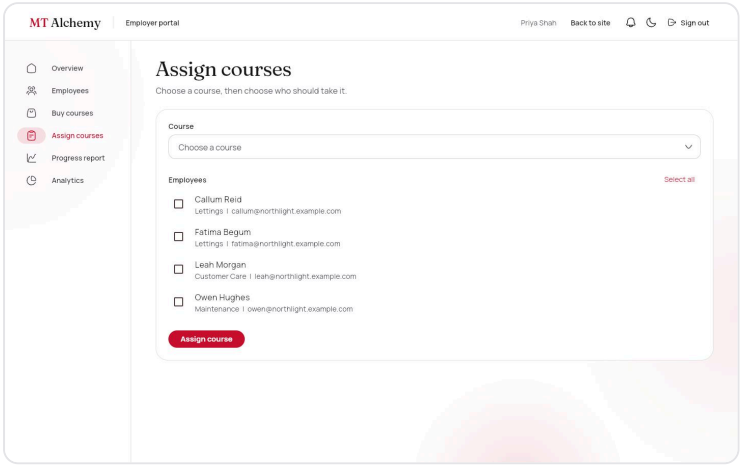
Every course as a photo card with a seat stepper and a Buy button that goes to checkout. Shows how many seats are unassigned.

DATA

courses, organizations.seats.

WHAT YOU CAN DO

- Choose seats
- Buy



41. Assign courses

`/portal/employer/assign`

Employer

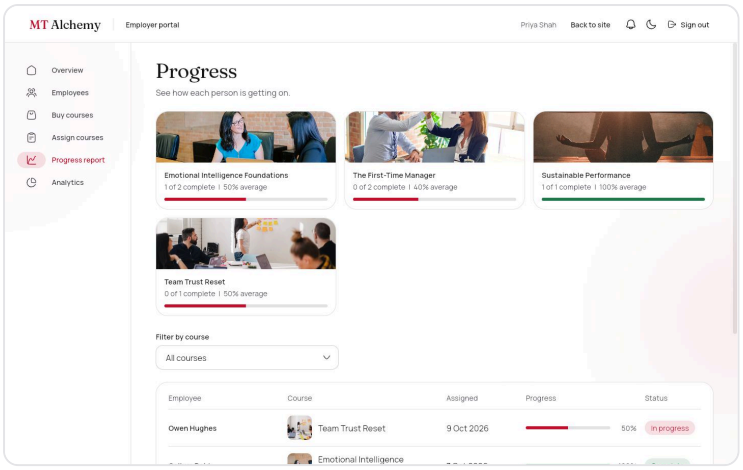
Choose a course, tick the people, assign. Shows the seats left and refuses if there are not enough, with a Buy seats shortcut.

DATA

assignments, organizations.seats.

WHAT YOU CAN DO

- Select all
- Assign
- Buy seats



42. Progress report

`/portal/employer/progress`

Employer

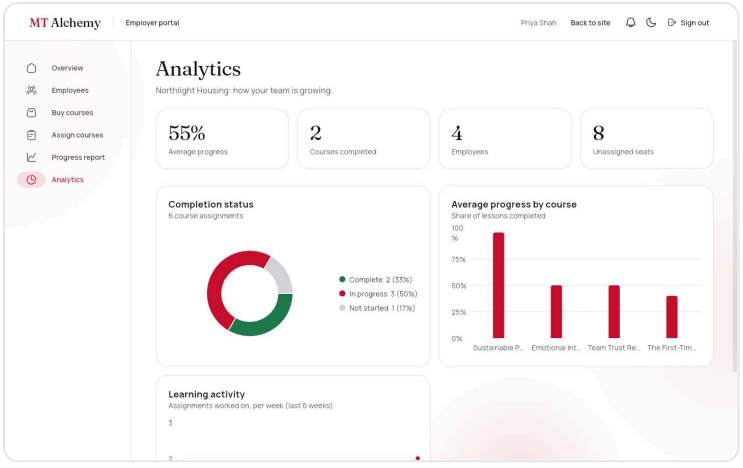
A photo summary card per course, then a filterable table of each person, course, date, progress bar and status.

DATA

assignments, enrolments (live).

WHAT YOU CAN DO

- Filter by course



43. Analytics

`/portal/employer/reports`

Employer

A completion donut, average progress by course (bars) and weekly learning activity (line), with hover details and a legend with counts.

DATA

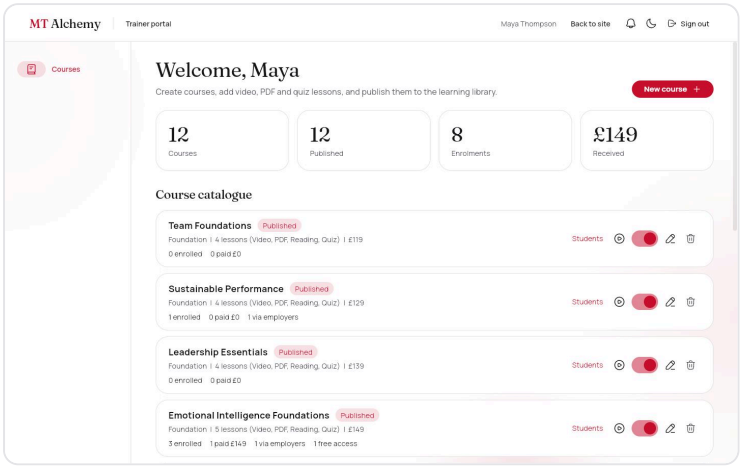
assignments, enrolments.

WHAT YOU CAN DO

- Hover for values

4.5 Trainer portal

For trainers, who author and look after courses. Super admins use the same screens.



44. Courses

`/portal/admin`

Trainer Super admin

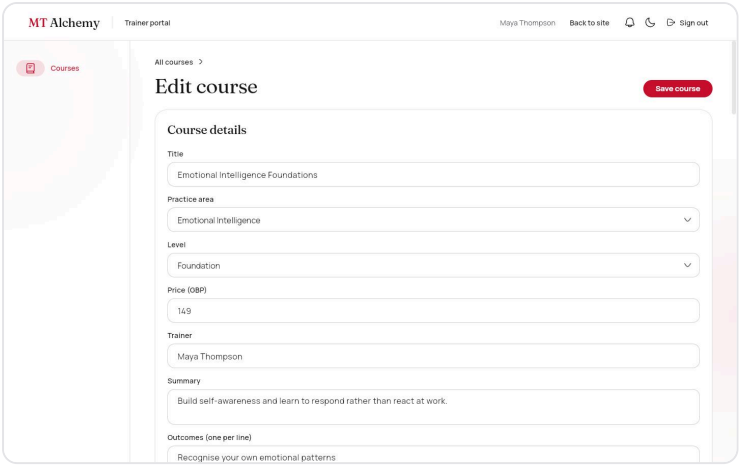
The whole catalogue with totals (courses, published, enrolments, amount received) and a row per course showing enrolled, paid with amount, employer seats and free access.

DATA

courses, enrolments (live).

WHAT YOU CAN DO

- Students
- Open free (play icon)
- Publish or unpublish
- Edit
- Delete
- New course



45. Course editor

`/portal/admin/courses/:id`

Trainer · Super admin

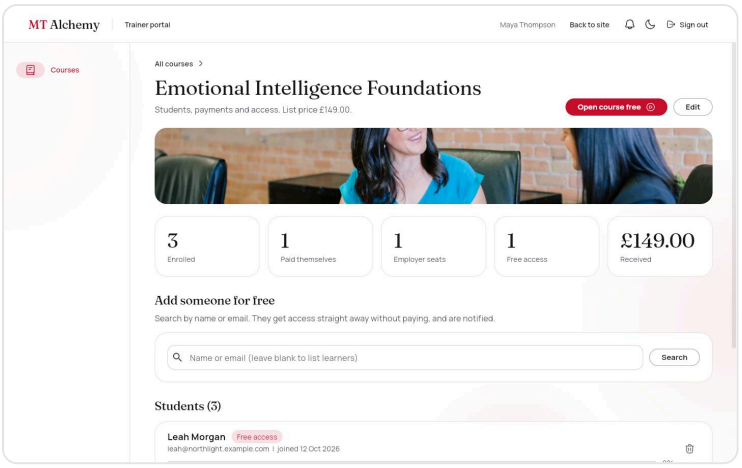
Create or edit a course: title, area, level, price, trainer, summary, outcomes, published switch, and any number of lessons (video, PDF, reading, quiz with questions and a correct answer), reorderable.

DATA

courses.

WHAT YOU CAN DO

- Add lessons by type
- Move up, remove
- Save



46. Students and payments

`/portal/admin/courses/:id/students`

Trainer · Super admin

Counts for enrolled, paid, employer seats, free access and amount received. Search any learner and give free access, or remove a student. Each row shows how they joined (Paid £149, Employer seat, Free access), their email, join date and progress.

DATA

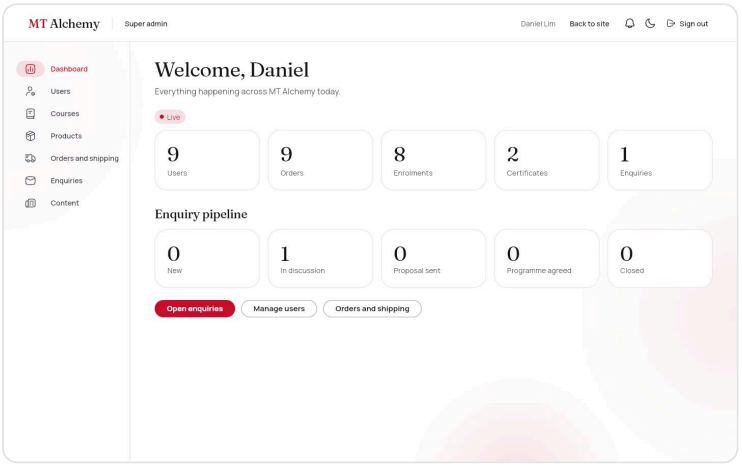
enrolments (live), users, notifications.

WHAT YOU CAN DO

- Search and Give free access
- Remove from course
- Open course free

4.6 Super admin portal

For the platform owner. Side navigation: Dashboard, Users, Courses, Products, Orders and shipping, Enquiries, Content.



47. Dashboard

`/portal/super`

Super admin

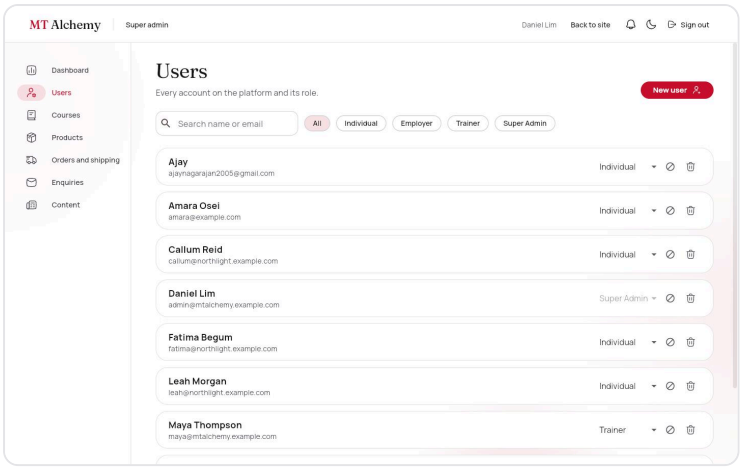
Live totals for users, orders, enrolments, certificates and enquiries (with a pulsing Live badge), and the enquiry pipeline by status.

DATA

Snapshot listeners on five collections.

WHAT YOU CAN DO

- Open enquiries, users, orders



48. Users

`/portal/super/users`

Super admin

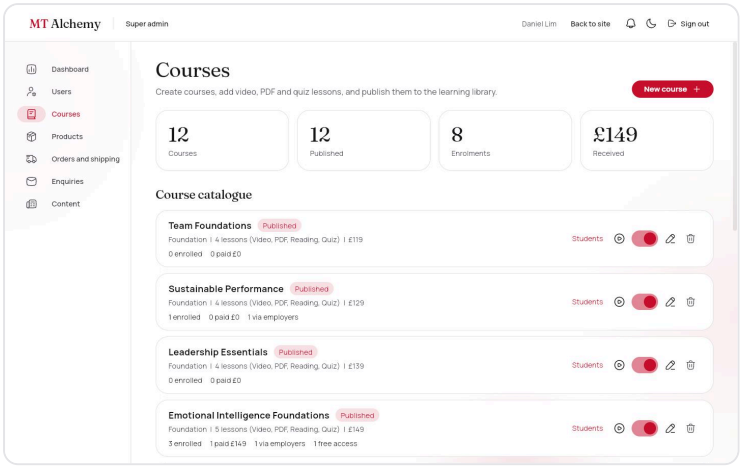
Every account with search and a role filter. Change a role, disable or enable, delete a profile, or create an account for anyone (using a second Firebase app so the admin stays signed in).

DATA

users.

WHAT YOU CAN DO

- Change role
- Disable
- Delete
- New user



49. Courses

`/portal/super/courses`

Super admin

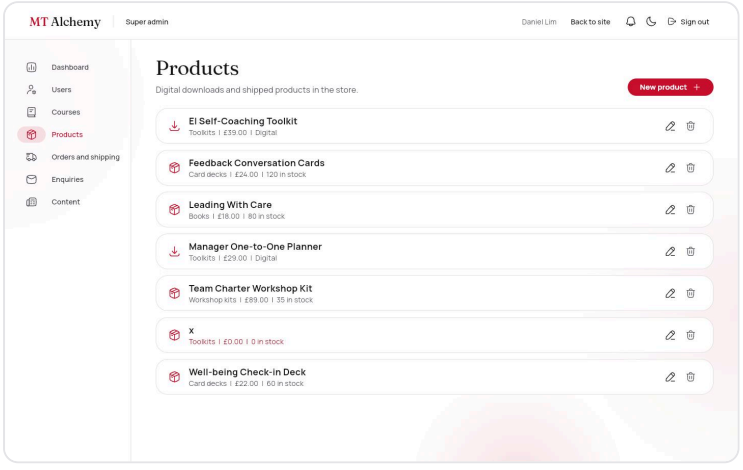
The same course management as the trainer portal, inside the super admin navigation.

DATA

courses, enrolments.

WHAT YOU CAN DO

- As the trainer portal



50. Products

`/portal/super/products`

Super admin

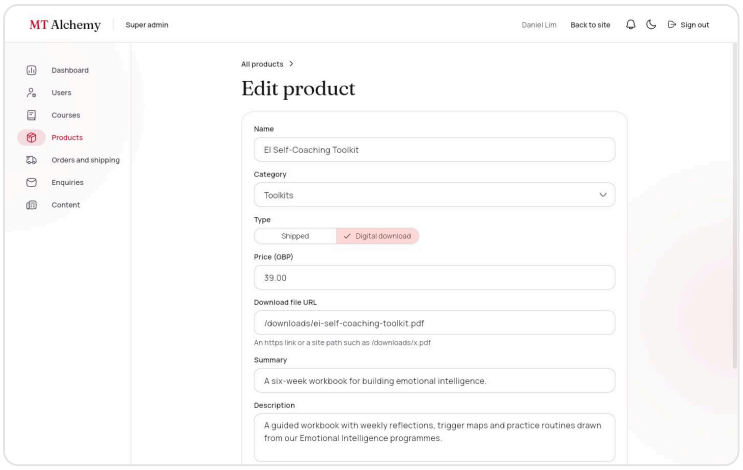
Digital and shipped products with category, price and stock (low stock shows in red).

DATA

products.

WHAT YOU CAN DO

- Edit
- Delete
- New product



51. Product editor

`/portal/super/products/:id`

Super admin

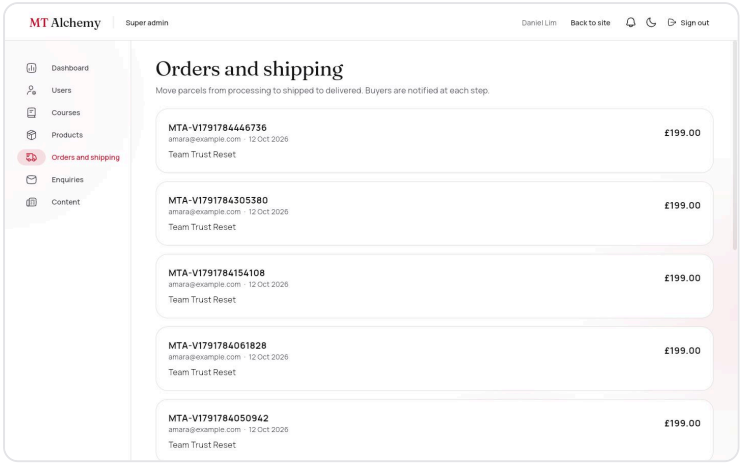
Name, category, type (Shipped or Digital download), price, stock or download file, summary, description and details.

DATA

products.

WHAT YOU CAN DO

- Save



52. Orders and shipping

`/portal/super/orders`

Super admin

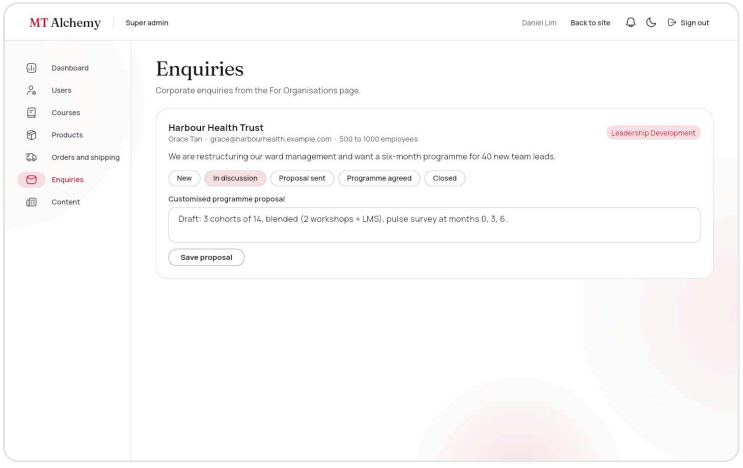
Every order with items, the delivery address and the delivery timeline. One button moves a parcel to the next status and notifies the buyer.

DATA

orders, shipping, notifications.

WHAT YOU CAN DO

- Mark as shipped
- Mark as delivered



53. Enquiries (Journey E)

`/portal/super/enquiries`

Super admin

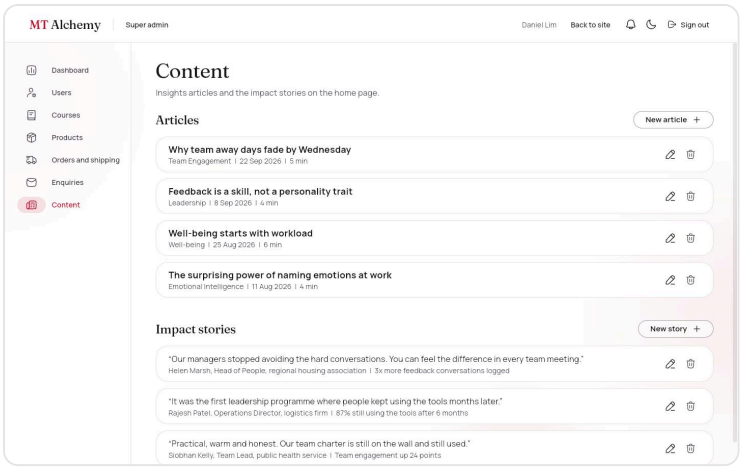
Corporate enquiries with a status pipeline (New, In discussion, Proposal sent, Programme agreed, Closed) and a field for the customised programme proposal.

DATA

enquiries.

WHAT YOU CAN DO

- Change status
- Save proposal



54. Content

`/portal/super/content`

Super admin

Create, edit and delete Insights articles and Impact Stories.

DATA

articles, impactStories.

WHAT YOU CAN DO

- New article, New story
- Edit, Delete

5. The five user journeys

A · Individual learning

1. Register: choose a career stage and interest areas (`/register`).
2. The course catalogue shows only courses that match both (`/portal/me/courses`).
3. Buy the course, then pay at checkout (test card 4242 4242 4242 4242).
4. An enrolment is created from the paid order; the dashboard shows progress.
5. Open the course: watch the video, read the PDF, Mark complete, Next.
6. Pass the quiz (70% or more) to complete the course and receive a certificate with a public verification link.

B · Digital product

1. Store, then the EI Self-Coaching Toolkit.
2. Add to cart, then checkout (no delivery address, no shipping).
3. A payment record is created in test mode.
4. My digital products, then Download. The count is tracked.

C · Physical product

1. Store, then Cart (delivery is free from £50 of shipped goods).
2. Enter a delivery address at checkout and pay.
3. Stock is reduced in a transaction; a shipment is created as Processing.
4. The super admin marks it Shipped (with a tracking number), then Delivered; the buyer is notified each time.

D · Employer

1. Register an organisation (`/register?type=organisation`).
2. Add employees; they join automatically when they register with that email.
3. Buy courses: choose the number of seats and pay.
4. Assign courses to people; each assignment uses a seat.
5. Watch the progress report and the analytics charts update live.

E · Corporate enquiry

1. Open "For organisations" from the landing page or footer.
2. Submit the enquiry (saved to Firestore).
3. The super admin inbox tracks it: New, In discussion, Proposal sent, Programme agreed, Closed.
4. The admin writes the customised programme proposal.

Staff: free access

1. A trainer or admin opens any course free (admin preview, nothing is saved).
2. Courses, then Students shows enrolled, paid, employer seats and free access.
3. Search a learner and give free access, or remove someone. A removed payer is not refunded automatically.

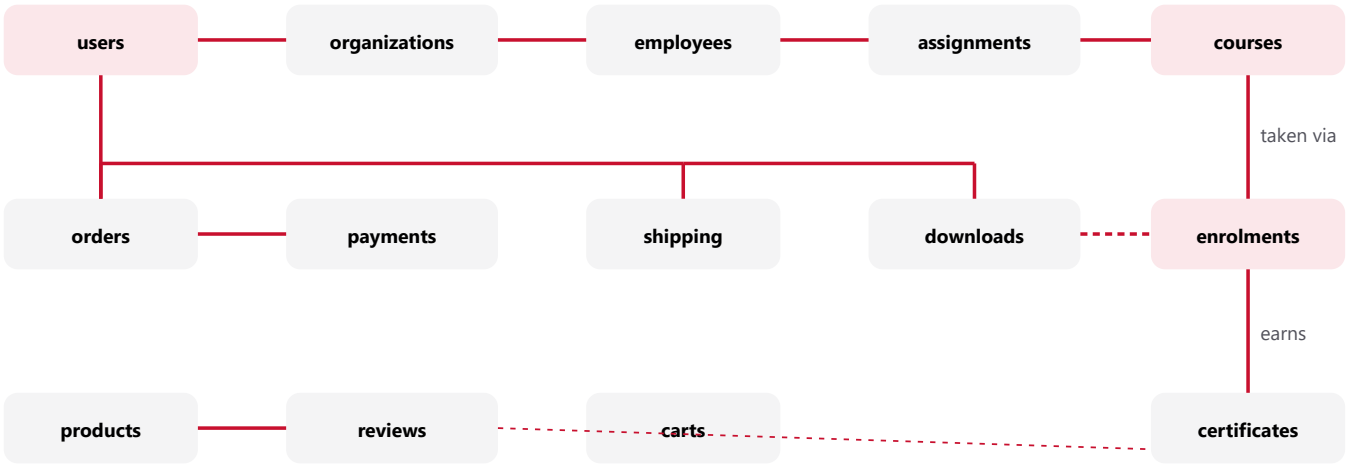
6. Data and security

Collections (22)

Each entity is a top-level collection. Lessons and quiz questions are embedded in the course document so the player loads in one read; order items are embedded in the order.

Collection	Key fields	Written by
users	name, email, role, orgId, careerStage, interests[], gender, country, hobbies[], disabled	The person at registration (individual or employer only); super admin
organizations	name, sector, size, ownerId, seats {courseId: n}	Employer
employees · assignments	orgId, name, email, team · employeeEmail, courseId	Employer
courses	title, level, serviceSlug, price, imageUrl, published, modules[] {type, mediaUrl, quiz[]}	Trainer, super admin
enrolments	userId, courseId, source (purchase, assignment, granted), amountPaid, orderId or assignmentId, completedModuleIds[], quizScores, progress, status	Learner (needs a paid order or an assignment); staff for free access
certificates	code, userId, courseId, userName, courseTitle, issuedAt (public read)	Learner, only when the enrolment is complete
products · productCategories	type (digital or physical), stock, fileUrl, imageUrl, price	Super admin (checkout may only reduce stock)
carts · orders · payments	items · status, totals, orgId · amount, last4, gateway "test-mode"	The buyer
shipping · downloads	status, tracking, history[] · url, downloadCount	Buyer creates; super admin advances shipping
articles · impactStories · founders	body[] · quote, metric · name, role, bio, imageUrl	Staff · super admin · super admin
reviews	targetId, userId, rating 1 to 5, comment	Signed-in learners
enquiries · contactMessages	organisation, interest, status, proposal · message	Anyone (guests too); the super admin manages them
notifications · assessments	userId, title, link, read · EQ scores	Staff and employers · the learner

Relationships



Also: articles · impactStories · founders · enquiries · contactMessages · notifications · assessments · productCategories (stand-alone)

Figure 6.1. Main relationships. Solid lines are foreign keys held in the child document; dashed lines are optional links (an enrolment may come from an order or an assignment). The full ER diagram is in `docs/ER_DIAGRAM.md`.

Access rules (from `firestore.rules`)

Collection	Guest	Individual	Employer	Trainer	Super admin
courses	read published	read published	read published	full CRUD	full CRUD
products, articles, stories, founders, reviews	read	read; write own reviews	read	read; articles CRUD	full
users	none	own profile (no role change)	own profile	read all	full
organizations, employees, assignments	none	own assignments	own organisation only	read	full
enrolments	none	own (needs paid order or assignment)	own organisation's	read; grant and remove free access	full
orders, payments, shipping, downloads	none	own	own organisation's orders	none	full; advances shipping
enquiries	create only	create only	create only	none	read and update

Key guarantees, each proven by the live verification script: users cannot change their own role; self-registration can only create individual or employer accounts; an enrolment needs a paid order or an assignment; a certificate can be created only when the enrolment is complete; employers cannot touch another organisation's data; draft courses are hidden from everyone except staff; buyers cannot mark their own parcel delivered.

Business rules

Rule	Behaviour
Learning pathway	A course is open to a learner only if its **level** suits their career stage **and** its **practice area** is one of their interests. Student: Foundation. Professional: Foundation and Practitioner. Manager: Practitioner and Leader. Senior leader: Leader. Other courses stay visible but locked, with the reason. Employer assignments and staff are not restricted.
Pricing and shipping	Prices in GBP. Delivery is £4.95, free from £50 of shipped goods. Digital products and courses never ship.
Seats	An employer buys seats per course (stored on the organisation). Assigning a course uses one seat per person; the assignment is refused when seats run out.
Progress and certificates	Progress is lessons completed divided by lessons. Quizzes pass at 70% and complete their lesson. Completing every lesson creates the certificate, whose code is built from the first three letters of the course id and the start of the learner id.
Payments	Simulated gateway in test mode. A card number starting 4000 0000 0000 0002 is declined; any other valid test number is approved. Only the last four digits are stored.
Free access	Staff can create an enrolment with source "granted"; it needs no order and no seat.
Offline fallback	If Firestore is unreachable the bundled catalogue keeps pages working.

7. Design system, imagery and motion

Colour

Pure white (#FFFFFF), a neutral zinc "ash" scale (#F4F4F5 to #18181B) and **Crimson Red #C8102E** as the single accent for calls to action, active states, progress and focus. Grey-to-red gradients appear only at moments of change: a passed quiz, a completed order, a finished course. There is no grain texture and no grey haze. A full dark palette mirrors the light one.

Glassmorphism

A shared `GlassPanel` (backdrop blur, translucent fill, hairline border) is used on the navigation, hero copy, summaries and forms. Few panels per screen, never nested, with a solid fallback for high contrast.

Typography

Fraunces (serif) for headings and **Manrope** for body text, with a fluid scale (display, h1 to h4, lead, body, small, label).

Imagery

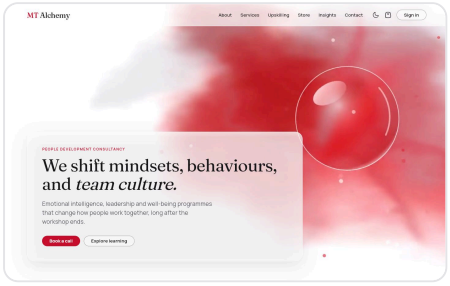
Thirty free-licence photographs (Unsplash, converted to WebP) appear on every course, product, service and article card. Courses and products store an `imageUrl` that admins can change; the seeded red ink artwork is the fallback. Founder portraits are uploaded from the site.

Motion

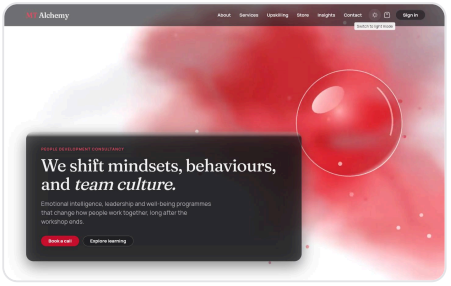
- Landing film with a floating glass crystal and parallax particles
- Staggered reveals on scroll
- Every card: hover lift, soft red shadow, photo zoom, press feedback
- 3D tilt on service cards
- Fade-and-lift page transitions
- Smooth glide from menu to landing section
- Animated EQ result bars, rotating impact stories
- All of it respects the reduced-motion setting

Accessibility

Semantic labels on icon buttons, live regions on form errors, visible focus, contrast-checked tokens, captions on the tour film, and an explicit table of text in every chart legend.



Light theme



Dark theme

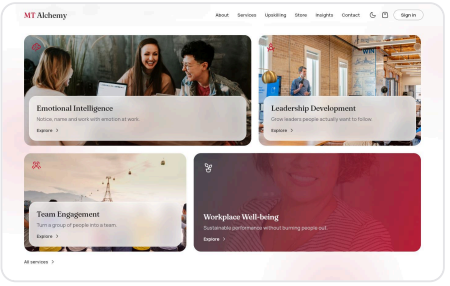


Photo service cards

Responsive behaviour

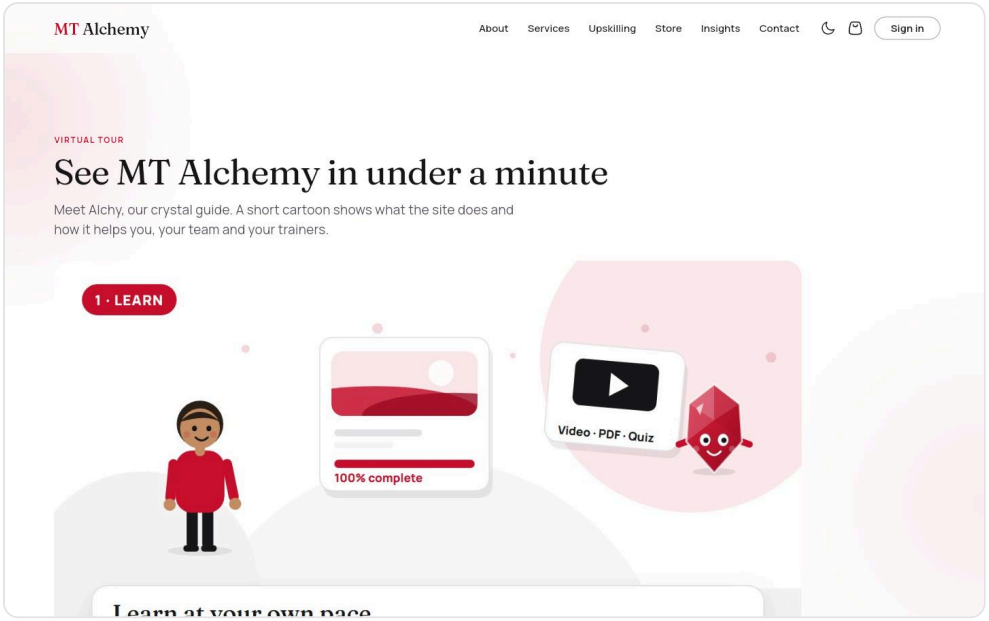
Size	Width	Layout
Compact	under 600 px	Single column, full-width buttons, banner photos on cards, drawers for navigation.
Medium	600 to 1023 px	Two columns, drawers for navigation.

Expanded	1024 px and up	Side rails, three or four columns, fluid gutters (5% of the width, 48 to 120 px).

8. Media pipeline: videos, virtual tour and QR code

All videos are made with **Remotion**, which renders React components to H.264 files, so every film is reproducible from code.

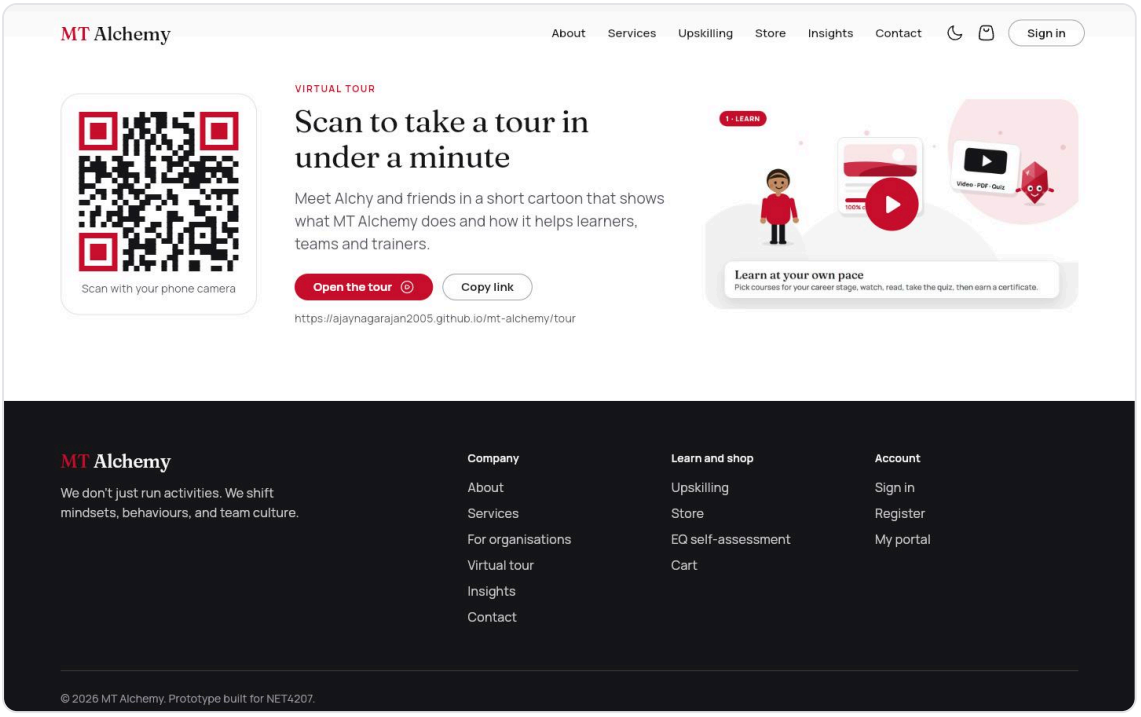
Film	Used on	Notes
`hero_loop` (+ dark)	Landing hero	Ink blooming through water, a floating glass crystal, parallax particles, a slow camera drift. Seamless 8 second loop, under 1 MB.
`services_loop` (+ dark)	Services header	Quiet ink loop.
`brand_intro`	About page, lesson video	Brand film with the wordmark and four practice areas.
`site_tour`	Virtual tour	42 seconds, 1280×720, no sound, with captions. Seven scenes: welcome, learn, shop, teams, trainers and admins, EQ check, call to action. Characters and the faceted-crystal mascot **Alchy** are drawn in SVG, so nothing is licensed.



The virtual tour page (`/tour`): the cartoon film, then one card per part of the site to try.

The QR code

The last section of the landing page shows a QR code that opens the tour. It encodes the site's own address, taken from the browser plus the build's base path, so it is correct on whatever host the site is published to. On a computer's `localhost` the page warns that the code only opens locally; on the live site it scans from any phone. The same section has an "Open the tour" button and a "Copy link" button.



QR section at the end of the landing page, with a preview of the film.

9. Quality and verification

Check	Result
`flutter analyze`	No issues.
`flutter test`	56 tests on a fake Firestore and mock Auth: cart rules; checkout writes (order, payment, shipping, download, enrolment, stock) and declined cards; shipping state changes with notifications; enrolment from an order and from an assignment; quiz pass and fail; certificate issue and verification; employer seats, assignments, progress and weekly activity; staff free access, removal and revenue totals; founders and photos; theme toggle; validators; portal scrolling; screen layout.
`npm run verify` (live Firebase)	39 of 39 checks pass. It signs in as each role over REST, performs each journey's writes, and confirms forbidden writes are refused, such as a learner promoting themselves, enrolling without paying, or giving themselves free access.
Visual checks	Browser screenshots at 1440 px and 375 px, light and dark; the live GitHub Pages site was checked including a deep link and a real sign-in.
Performance	Videos under 1.5 MB each; photographs 25 to 170 KB (WebP); the live loop video and 3D crystal load lazily.

Lesson from testing. Firestore rule changes take a few minutes to reach every server. A verification run started during that window briefly saw the temporary open seed rules. The seed and maintenance notes now say to wait about three minutes before verifying, and the verification script cleans up the records it creates.

10. Deployment and operations

Task	How
Run locally	`flutter pub get` then `flutter run -d chrome`
Publish the site	Commit, then `bash tools/deploy_pages.sh`. It builds with the `/mt-alchemy/` base path, copies `index.html` to `404.html` (so deep links such as `/portal` load), adds `.nojekyll` and pushes the `gh-pages` branch.
Change rules or indexes	Edit `firestore.rules` or `firestore.indexes.json`, then `firebase deploy --only firestore`, wait three minutes, then `npm run verify`.
Reset demo data	`cd tools/seed && npm run seed` (exports the catalogue, opens the temporary seed rules, writes 71 documents, redeploys the real rules).
Re-render a video	`cd video && npm run render:tour` (or `render:all`).
Authorized domains	The Firebase Authentication allow-list contains `localhost`, `*.firebaseapp.com`, `*.web.app` and `ajaynagarajan2005.github.io`. Add any new domain under Authentication, Settings.
Day-to-day content	Trainer portal (courses), Super admin (users, products, orders, enquiries, content), landing page founder cards (photo upload for the super admin).

Troubleshooting

Symptom	Likely cause and fix
"You do not have permission to do that."	A security rule refused the write. Check the role in `users/{uid}` and reproduce it in `verify.mjs`.
Catalogue shows but never updates	Firestore unreachable; the app falls back to the bundled catalogue. Look for "Could not reach Cloud Firestore" in the console.
"requires an index"	A new compound query. Follow the link in the error or add it to `firestore.indexes.json`.
Portal is empty after login	The profile document is missing. Re-run the seed or create `users/{uid}` with a role.
QR code opens a blank or wrong address	The build was made without `SITE_BASE`; rebuild with `deploy_pages.sh`.

11. Limitations and future work

Area	Current state and the next step
Payments	Simulated in test mode, with no real charge. Next: a payment provider (Stripe or similar) with a server-side webhook, which needs Cloud Functions.
File storage	Firebase Storage needs the paid plan, so PDFs and downloads are served from the site's own `/downloads` folder, and founder photos are stored in Firestore as small JPEGs. Next: Storage for lesson videos, PDFs and photos.
Email	Password-reset email is sent by Firebase. Order receipts, employee invitations and enquiry replies are not emailed; people see them in the app (notification bell). Next: transactional email.
Certificates	Viewable and verifiable by link; no PDF download button yet.
Refunds	Removing a paying student does not refund them; an admin must handle that with the payment provider.
Devices	Verified in desktop and phone-sized browsers. Native Android and iOS builds use the same code but were not tested on devices.
Browser window fit	One report of the page not filling the window edge to edge on a particular display was not reproduced and remains open.
Hosting deep links	GitHub Pages returns HTTP 404 for deep links before the app loads (the app then renders normally). Firebase Hosting would return 200.
EQ self-assessment	An engagement tool with ten statements across five areas, not a validated psychometric instrument.
Hero video	An AI-generated landing clip was deferred; the Remotion films are used instead.

12. Appendices

A. Demo accounts

Password for every account: `alchemy123`. The login page also lists them.

Role	Email	What to try
Individual	amara@example.com	Course in progress, a shipped order, a digital download, pathway: Professional with three interest areas
Employee (individual)	callum@northlight.example.com	Courses assigned by an employer, pathway: Manager
Employer	priya@northlight.example.com	Employees, seats, assignments, progress, analytics
Trainer	maya@mtalchemy.example.com	Course CRUD, open courses free, students and free access
Super admin	admin@mtalchemy.example.com	Everything: users, products, orders, enquiries, content, founders' photos

Test card for checkout: `4242 4242 4242 4242` (approved) or `4000 0000 0000 0002` (declined); any future expiry and any 3-digit CVC.

B. Route table

Route	Screen	Access
`/`	Landing page	Public
`/about` · `/services` · `/services/:slug`	About; services; service detail	Public
`/upskilling` · `/upskilling/:id`	Course catalogue; course detail	Public
`/learn/:id`	Course player	Signed in
`/store` · `/store/:id`	Store; product	Public
`/cart` · `/checkout` · `/checkout/success`	Cart; checkout; confirmation	Cart public; checkout signed in
`/insights` · `/insights/:slug`	Articles	Public
`/contact` · `/organisations`	Contact form; corporate enquiry	Public
`/eq-assessment` · `/tour` · `/certificates/:code`	EQ check; virtual tour; certificate	Public
`/login` · `/register`	Sign in; register (individual or organisation)	Public
`/portal/me` and `/courses` `/certificates` `/orders` `/downloads` `/profile`	Learner portal	Individual

`/portal/employer` and `/employees` `/courses` `/assign` `/progress` `/reports`	Employer portal	Employer
`/portal/admin`, `/portal/admin/courses/:id`, `…/students`	Trainer courses; editor; students	Trainer, super admin
`/portal/super` and `/users` `/courses` `/products` `/products/:id` `/orders` `/enquiries` `/content`	Super admin portal	Super admin

C. Credits and licences

- **Photographs:** free-licence images from Unsplash (people, workshops, well-being, books, teams), converted to WebP. The Unsplash licence permits this use without attribution.
- **Fonts:** Fraunces and Manrope (SIL Open Font Licence) via Google Fonts.
- **Icons:** HugeIcons (stroke rounded set).
- **Cartoon characters and mascot:** original artwork drawn for this project.
- **Packages:** Flutter, go_router, provider, Firebase plugins, fl_chart, qr_flutter, video_player, file_picker, image, flutter_animate, Remotion and their dependencies, each under its own open-source licence.

D. Glossary

Pathway	A learner's career stage plus interest areas; decides which courses they can buy.
Seat	One employee's place on a course, bought by an employer and used when assigning.
Free access (granted)	An enrolment created by staff with no payment.
Admin preview	Staff opening a course without enrolling; nothing is saved.
Alchy	The crystal mascot who narrates the virtual tour.
Seed	The script that fills Firebase with demo accounts and content.
Test mode	Payments are simulated; no real card is charged.